

An introduction to protein structure, folding and function

Main sources:

Introduction to protein structure
by Carl Branden and John Tooze

Protein structure and function
by A. Petsko and Dagmar Ringe

Molecular Biology of the cell
by Alberts:
Chapter 3 The atomic structure of proteins
Self study : Protein Function

Titia Sixma
t.sixma@nki.nl

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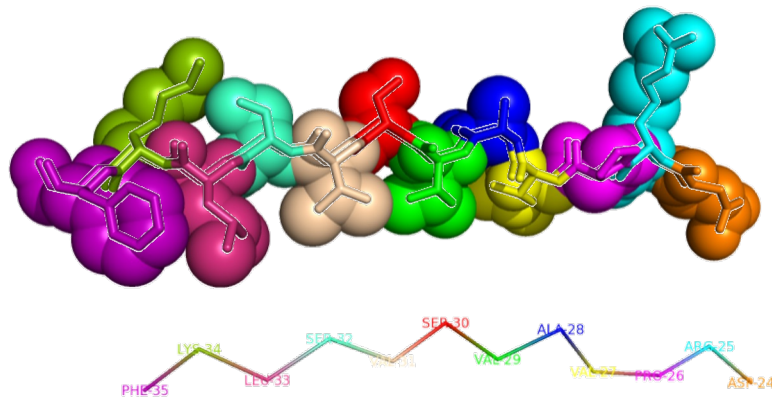
Proteins regulate the cell

- Regulate and execute cellular processes
 - Understand when
 - Understand who
 - Understand how
 - Mechanistic insight
 - Search for ability to interfere (Drug design)

Lay a basis for understanding the role that proteins play in cellular processes

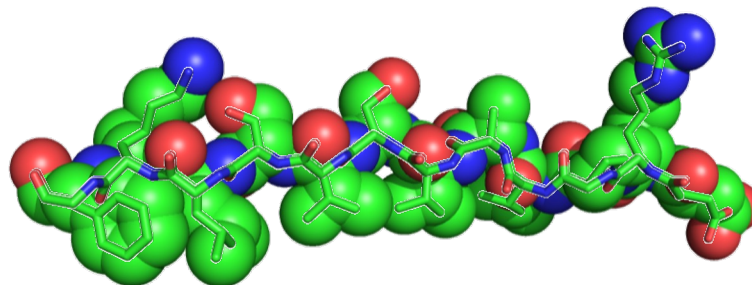
2

Proteins are chains of amino acids



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Atom types determine properties

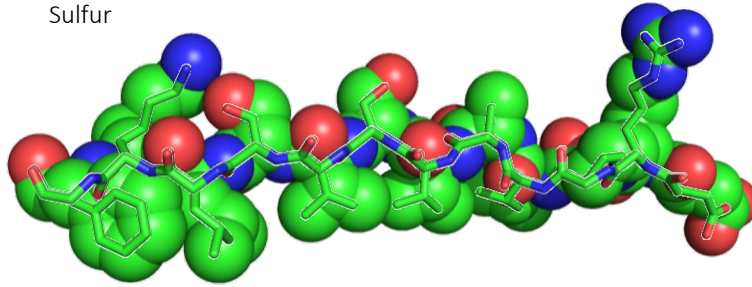


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Atom types determine properties

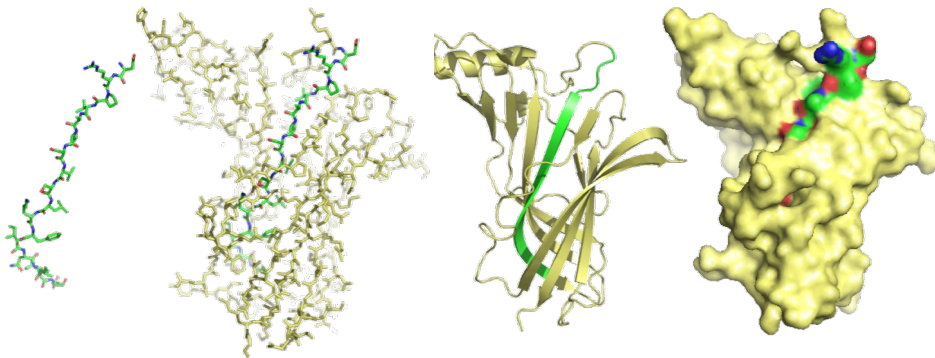
The 20 different aminoacids consist of

- Carbon (green)
- Nitrogen (blue)
- Oxygen (red)
- Sulfur



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The atomic structure of proteins



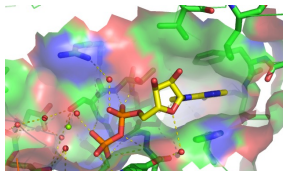
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Topics for this lecture

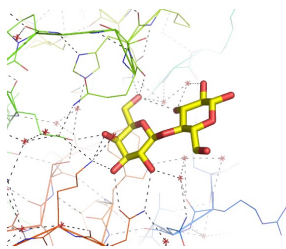
- Basic principles
 - Interactions covalent and noncovalent
 - The hydrophobic effect
- Protein structure
 - Primary, secondary, tertiary, quaternary
- Protein folding
 - Theoretical, in vitro, in vivo, prediction
- Protein function
 - Enzyme
 - Structural
 - Signalling
 - Motor activity

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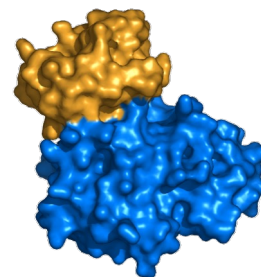
Protein function is always through interactions



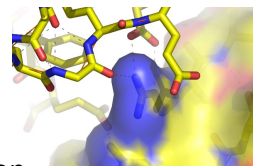
Protein-ligand interaction



Protein-DNA interaction



Protein-protein interaction



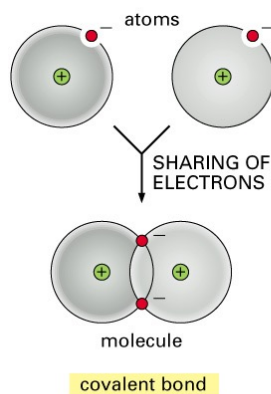
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Let's talk about interactions

- Covalent bonds
 - Along the amino acid sequence: peptide bonds
 - Disulfide bridges
- Non-covalent bonds
 - *Salt bridges (ionic bonds)*
 - Hydrogen bonds
 - Vanderwaals interactions: nonpolar interaction
- The hydrophobic effect

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Covalent bonds



Ionic bonds in vacuum covalent, in water noncovalent

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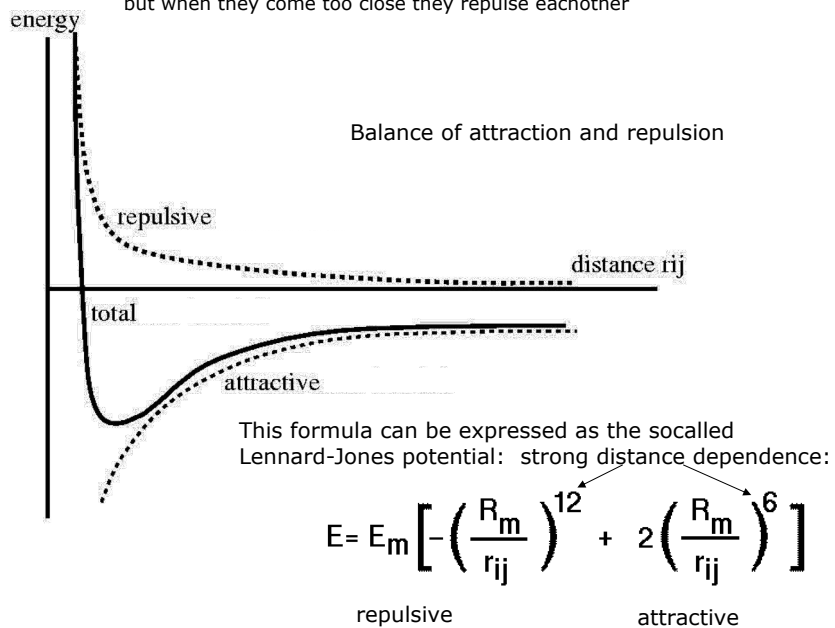
Non-covalent interaction

- Electrostatic : ionic bonds
 - Between opposite charges: attractive
 - Between identical charges: repulsive
- Van der Waals contacts (~ 0.2 kcal/mol)
 - Close to atoms: repulsive ($1/r^{12}$)
 - Slightly further: attractive ($1/r^6$)
- Hydrogen bonds ($\sim 2-5$ kcal/mol)
 - Special case, almost **covalent** character
 - Hydrogen bond donors: NH, OH
 - Hydrogen bond acceptors: O, N

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Van der Waals forces

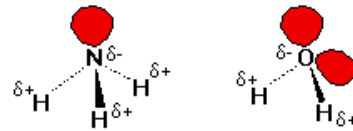
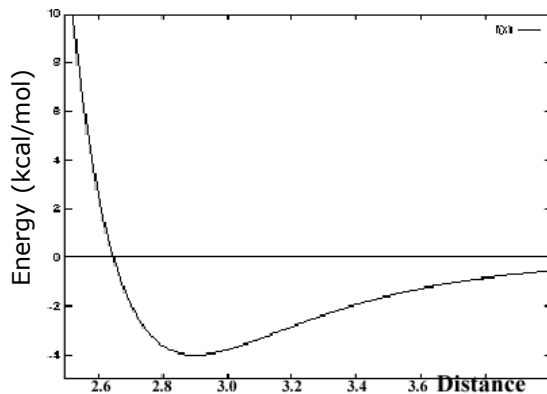
When two atoms approach each other they undergo a weak attractive force, but when they come too close they repulse each other



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Non covalent interactions: the hydrogen bond

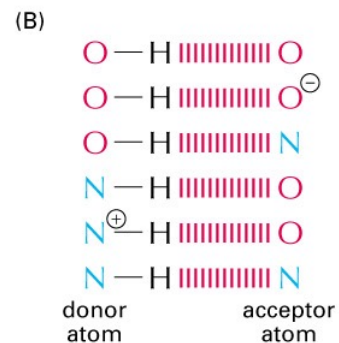
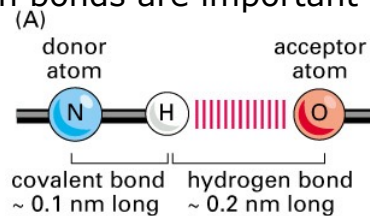
Hydrogen bonds are almost 'weak' covalent bond
 Short optimal distance (2.8 Å)
 Directional component



Hydrogen bond potential

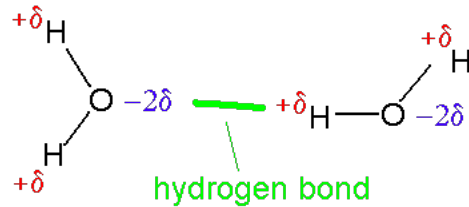
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Hydrogen bonds are important non-covalent bonds



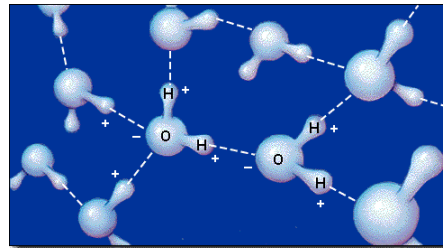
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The hydrophobic effect: Water makes oriented hydrogen bonds;



Water makes many hydrogen bonds
Water is oriented by hydrogen bonds
Non-polar amino acids prevent ideal hydrogen bonds in surrounding water
This causes the water to be more ordered: unfavorable

It is more favorable to sequester the apolar (hydrophobic) amino acids away from the water



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Information about detailed atomic interactions

- **Van der Waals contacts** (~0.2 kcal/mol)
 - Close to atoms: repulsive ($1/r^{12}$)
 - Slightly further: attractive ($1/r^6$)
- **Hydrogen bonds** (~2-5 kcal/mol)
 - Special case, almost **covalent** character
 - Hydrogen bond donors: NH, OH
 - Hydrogen bond acceptors: O, N
- **Electrostatic (ion pair)**
 - Between opposite charges: attractive
 - Between identical charges: repulsive

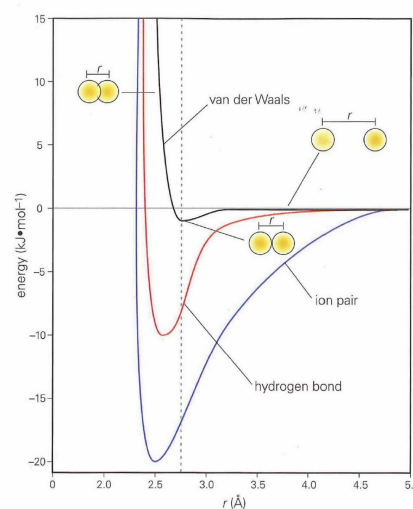


Figure from: Kuriyan, The Molecules of Life

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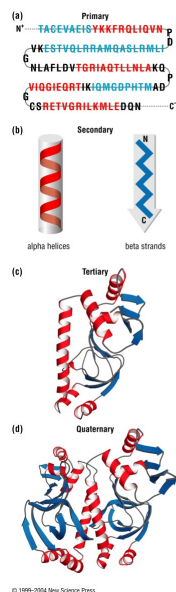
Interactions

	Length nm	Strength (H2O) kcal/mole
• Covalent	0.15	90
• Noncovalent		
– Ionic bond (salt bridge)	0.25	3
– Hydrogen bond	0.30	1
– Van der Waals	0.35	0.1

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Protein folding

From *Protein Structure and Function*
by Gregory A Petsko and Dagmar Ringe



Primary



Secondary

polypeptide chain of
amino acids

alpha-helices,
beta-sheets

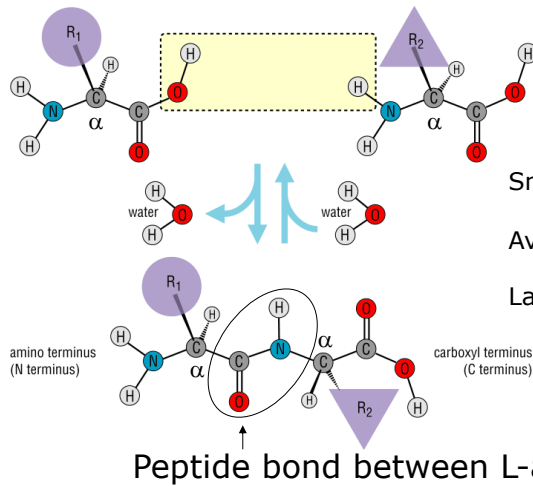
Tertiary

domain fold

Quaternary relative orientation
of domains and
subunits

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A protein is a covalent chain of amino acids, linked by peptide bonds



Smallest proteins < 100 a.a.

Average proteins ~ 300-500 a.a.

Largest proteins > 3000 a.a.

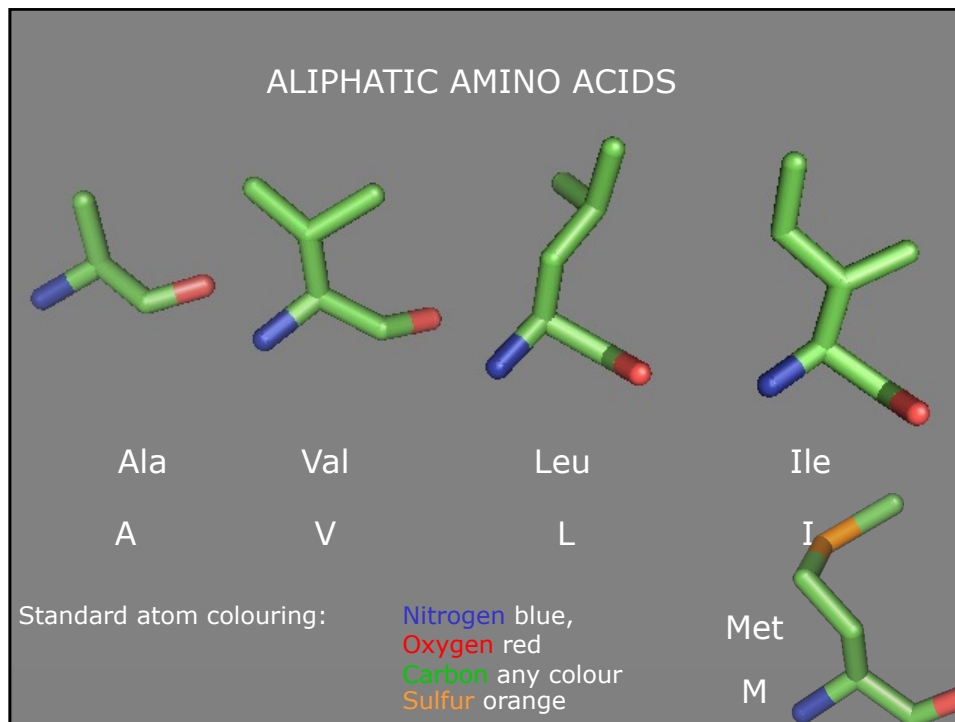
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There are 20 natural amino acids

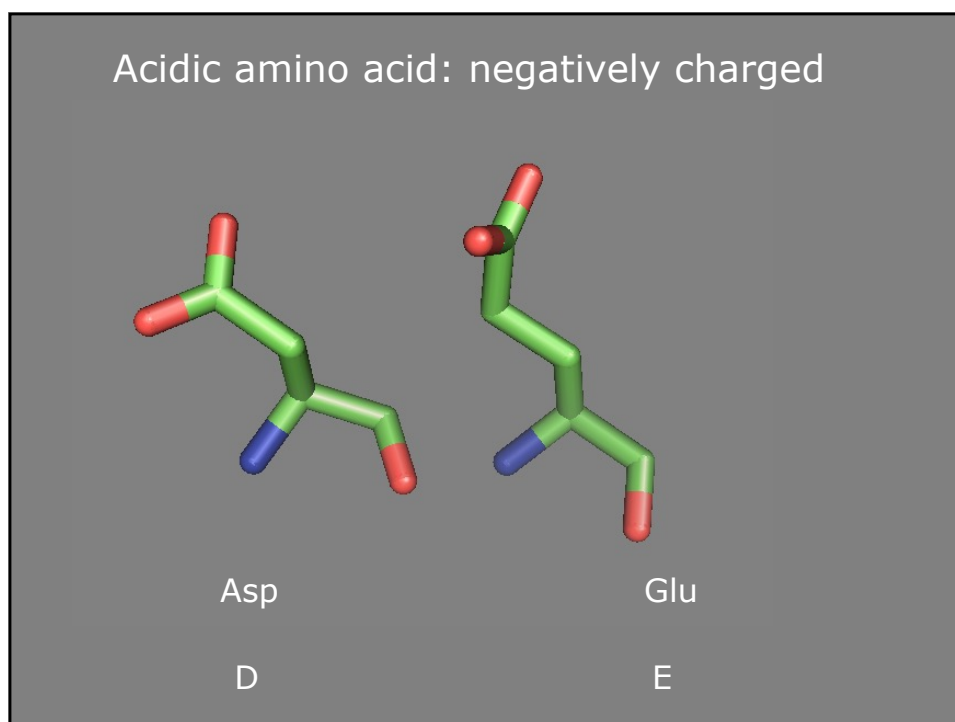
Amino acids can be classified into 5 groups dependent on side chain

1. **Aliphatic:** R = hydrocarbon side-chain
2. **Charged:** R carries charge in solution at pH 7
positive (basic) or negative (acidic)
3. **Aromatic:** R = aromatic ring
4. **Polar:** R = uncharged, polar group
5. **Conformational**

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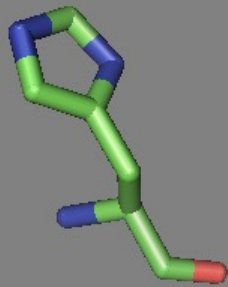


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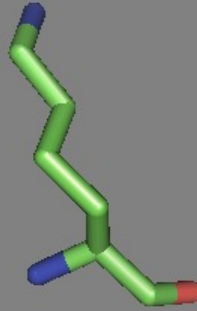
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Basic amino acids: positively charged



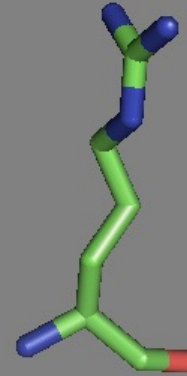
His

H



Lys

K

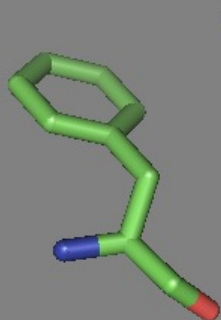


Arg

R

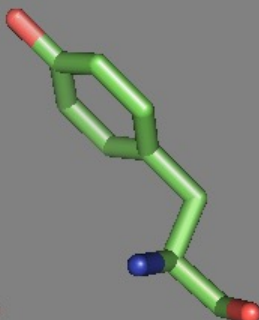
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Aromatic amino acids



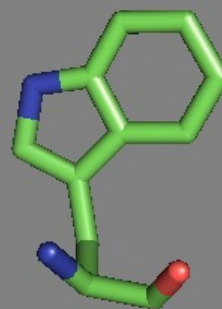
Phe

F



Tyr

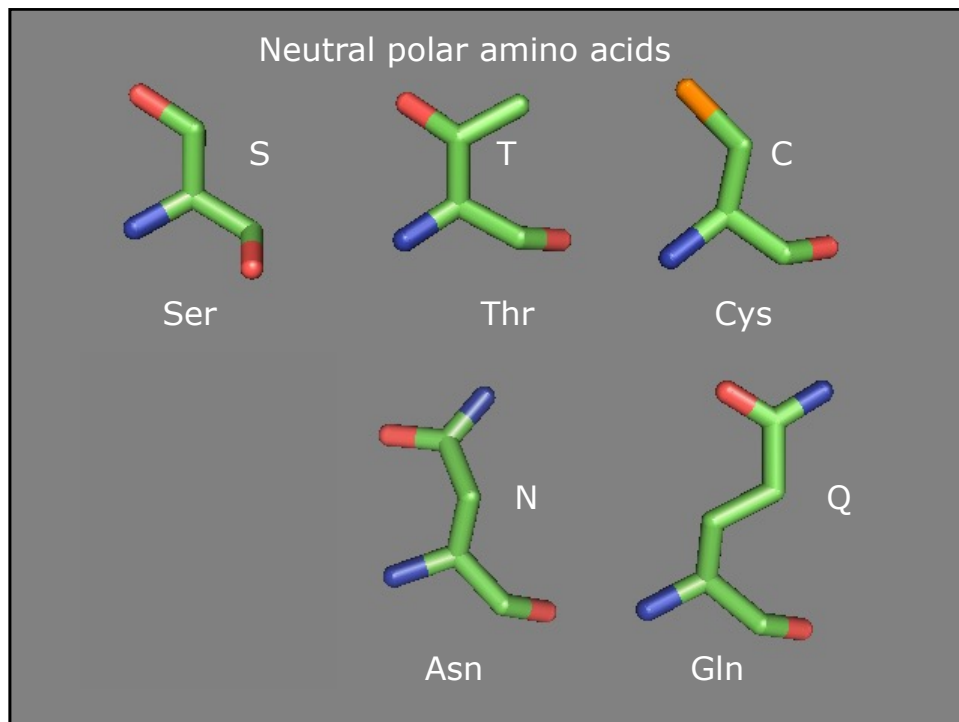
Y



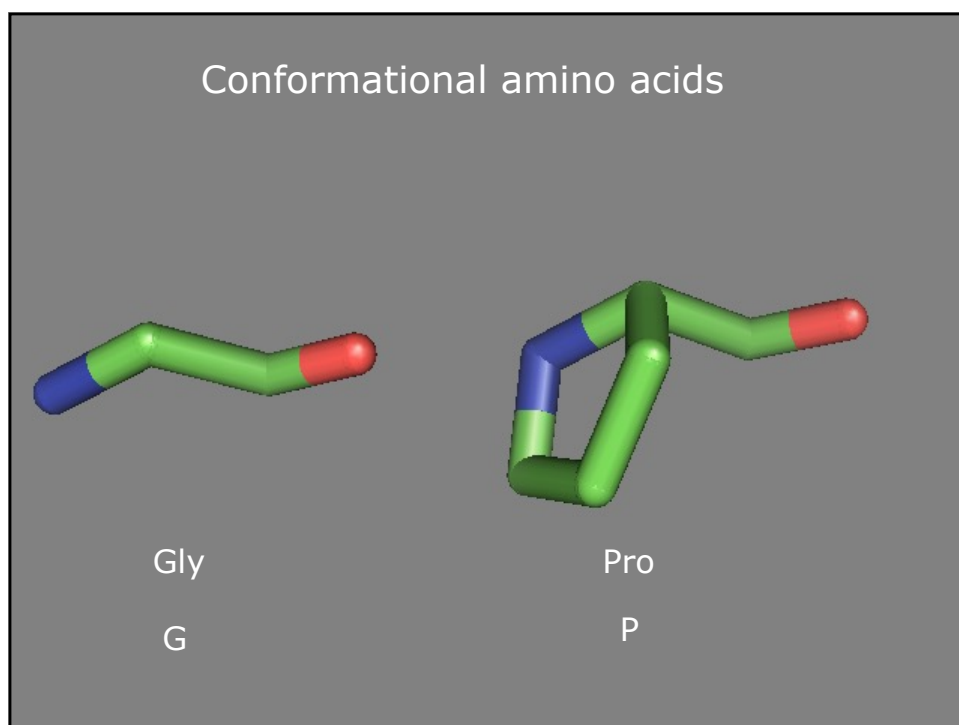
Trp

W

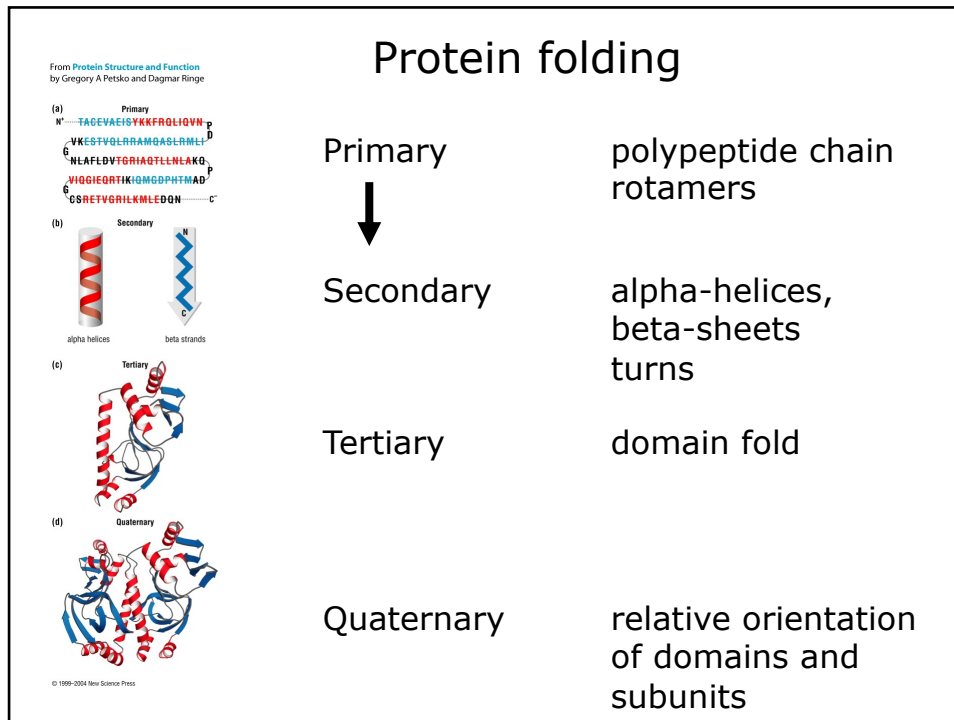
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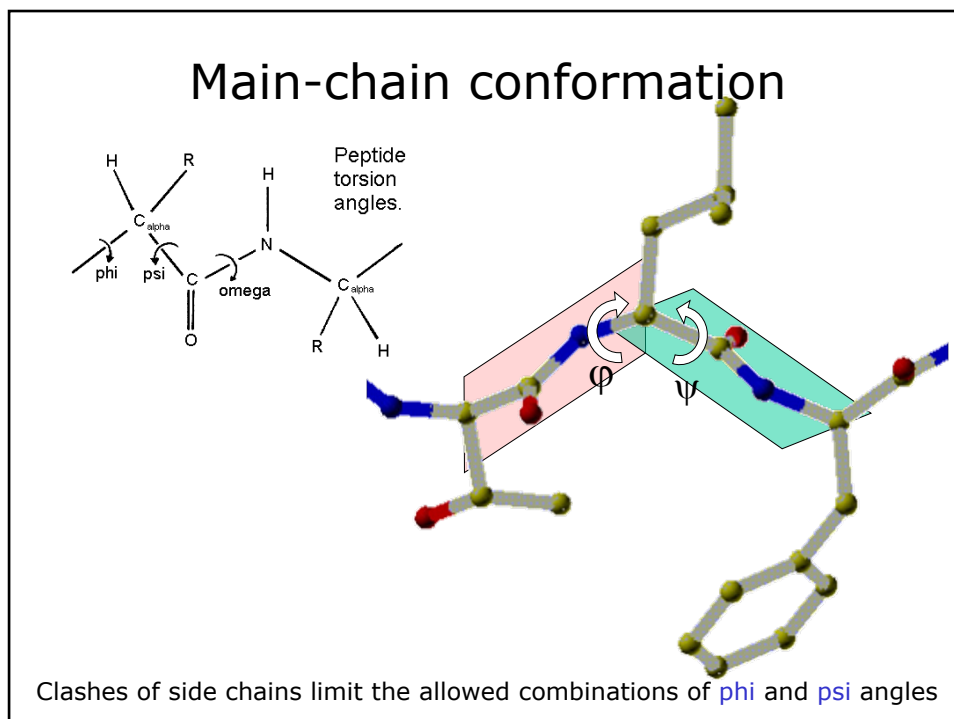
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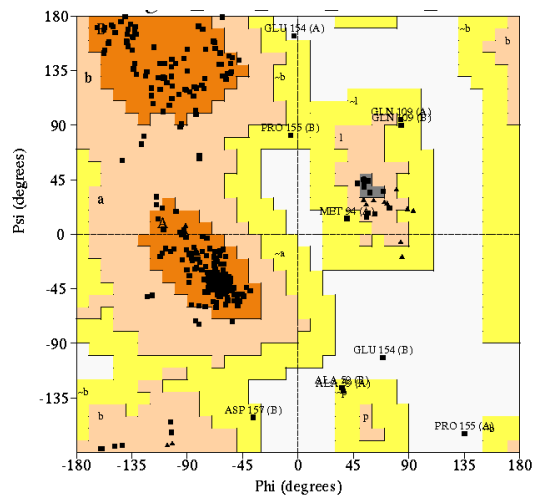


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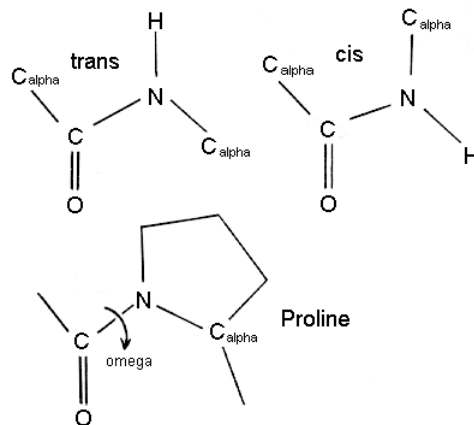
Ramachandran plot



Ramachandran plot shows main chain Phi/Psi combinations plotted in regions of 'allowed' (orange) and 'forbidden' (white) combinations. Occasionally a 'strained' forbidden combination is found in active sites of proteins.

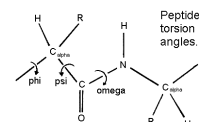
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The peptide bond usually adopts the trans conformation



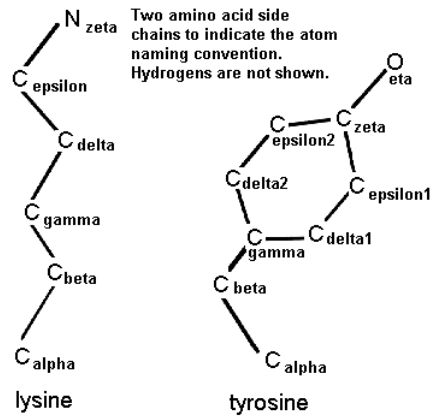
Rotation about omega angle

Proline relatively often in cis conformation



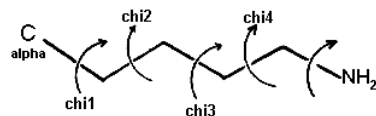
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Side chain nomenclature



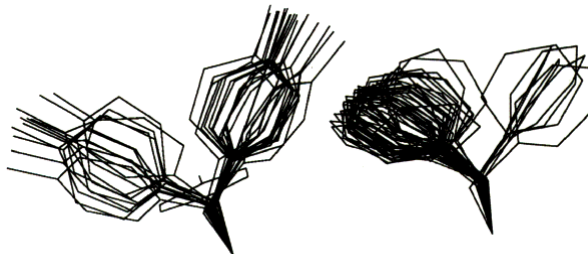
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Amino acid side chains adopt preferred conformations or 'rotamers'



Side chain conformation:

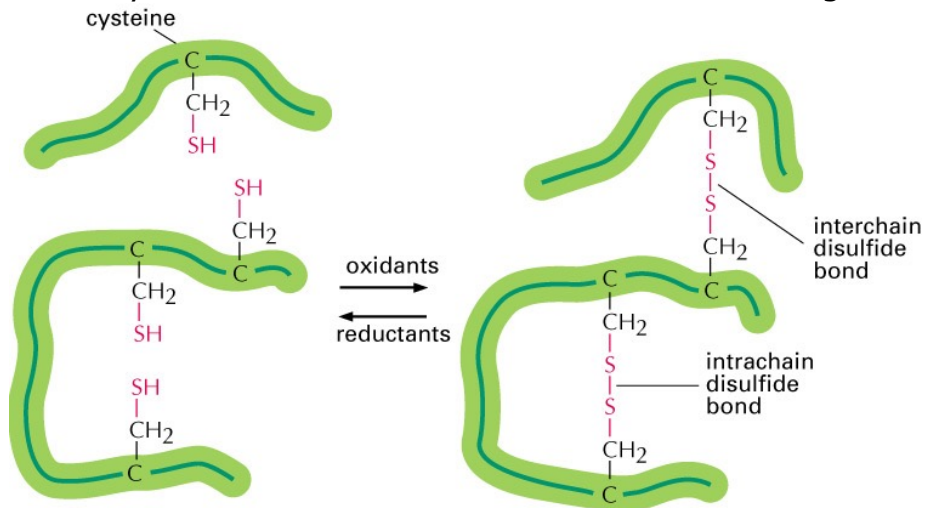
- Expressed in torsion angles Chi1, Chi2 etc.
- Steric hindrance with main chain causes preferred combinations: rotamers



Picture:
<http://xray.bmc.uu.se/~lars/Practicals/HomMod/rotamer%20distribution3>

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Two cysteines can make a covalent disulfide bridge

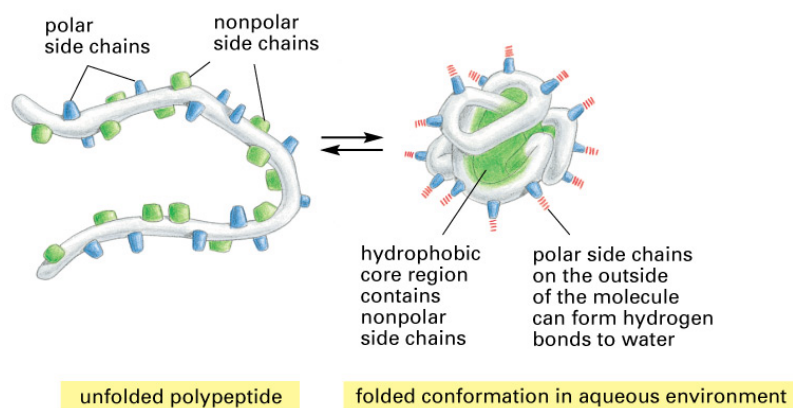


Disulfide bridges require oxidizing conditions, and therefore are not normally found in the cytosol: Normally only in the secretion route of cells or extracellular

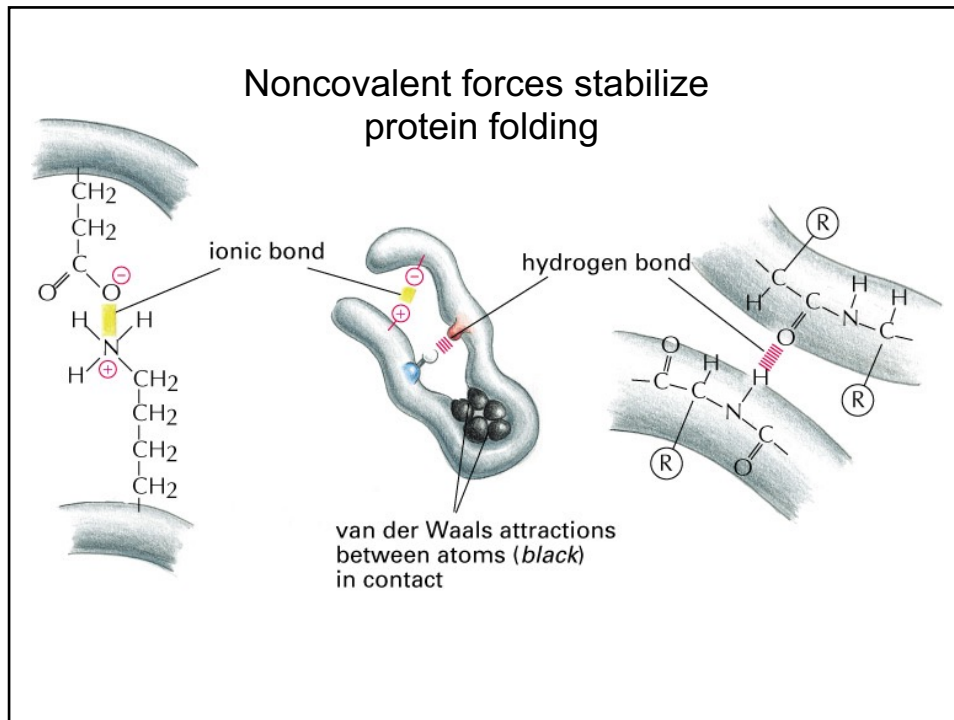
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Folding

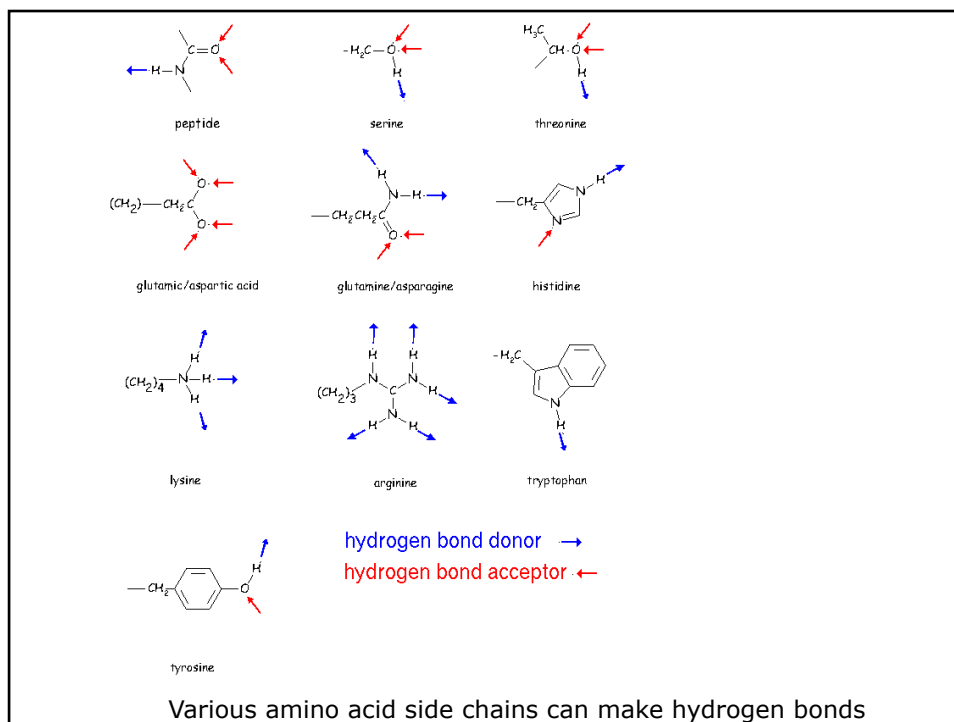
Proteins do not form extended chains in water
They behave like oil drops in water



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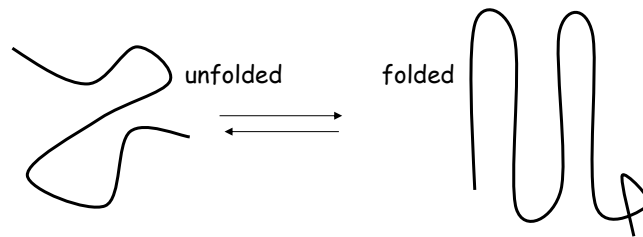
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Protein folding

Energy difference between denatured and folded state: 5-15 kcal/mol

Energy difference for 1 hydrogen bond: 2-5 kcal/mol

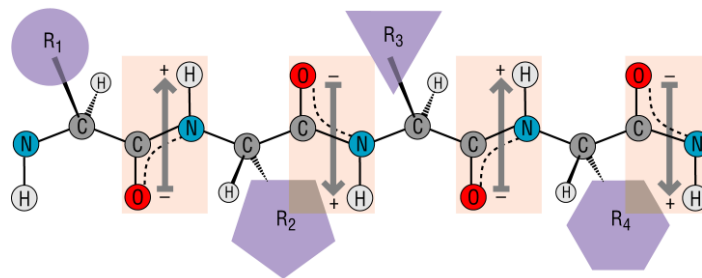
What drives protein folding?



- Prevention of the ordering of water, by sequestering hydrophobic residues
- Making sure that all possible hydrogen bonds are made after folding

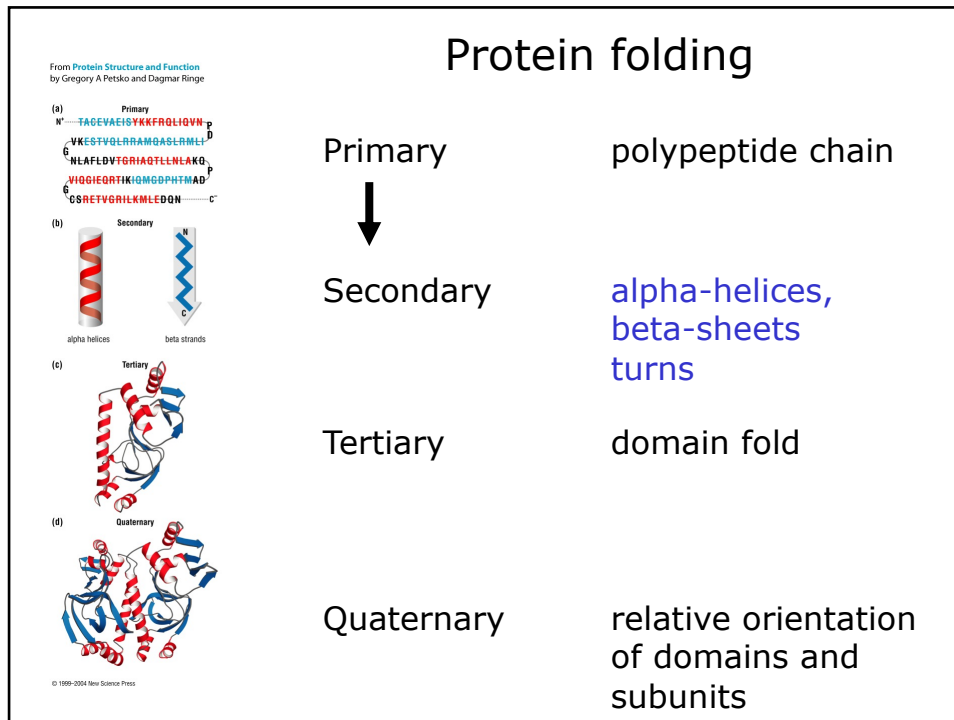
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Problem ? The main chain is hydrophilic!

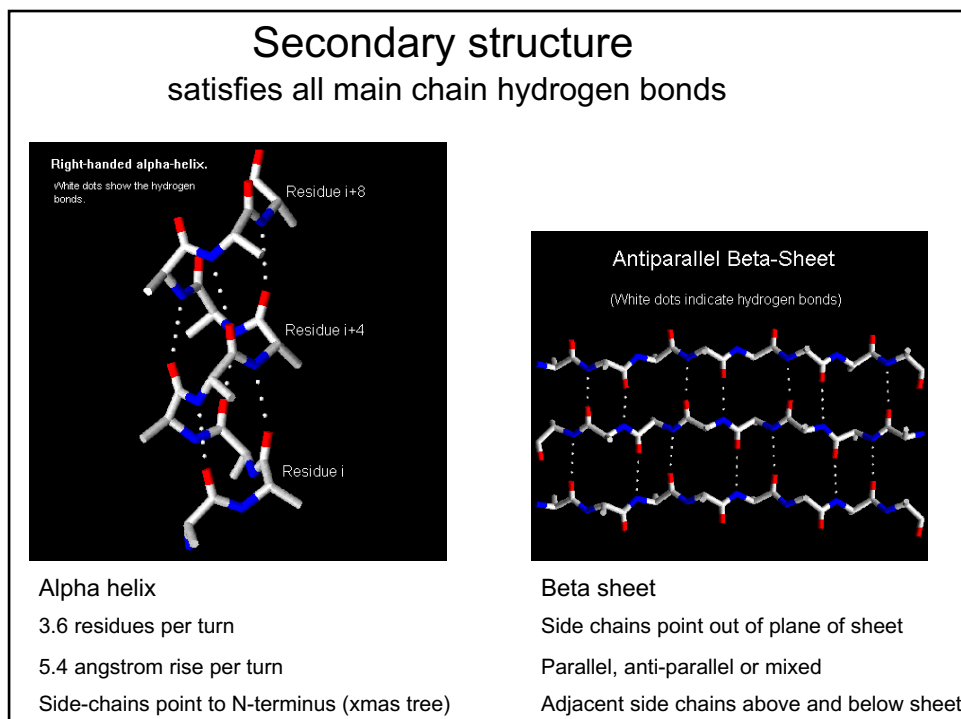


Solution:
make all possible hydrogen bonds by creating
'secondary structure'

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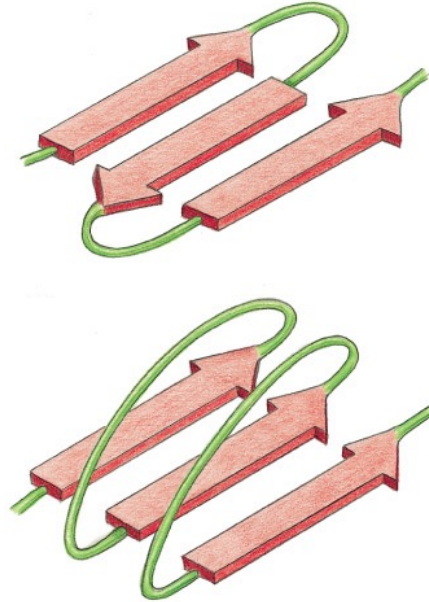


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Beta sheets can be parallel or anti-parallel



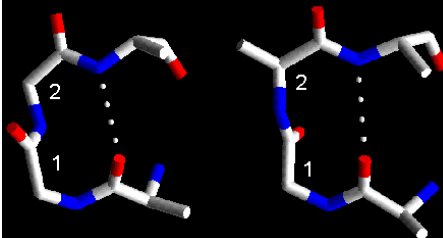
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Secondary structure elements are connected by turns and loops

Two-residue beta-hairpin turns.

Type I'

Type II'



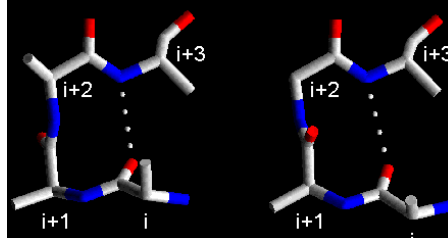
White dots indicate hydrogen bonds.

The main difference between these two turns is the orientation of the peptide group between residues 1 and 2.

Reverse turns.

Type I

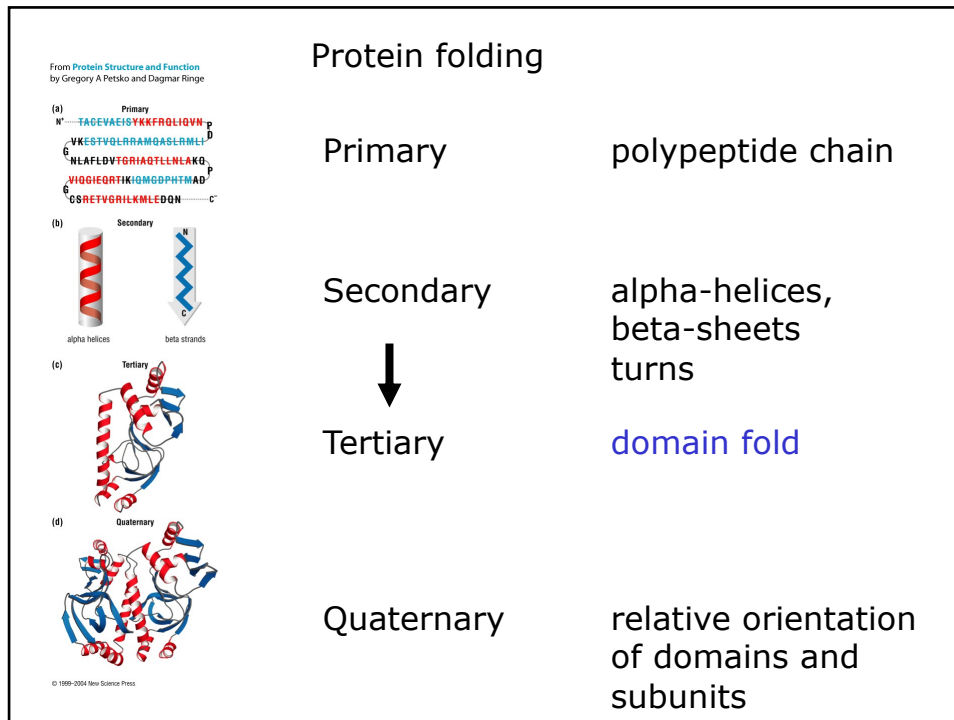
Type II



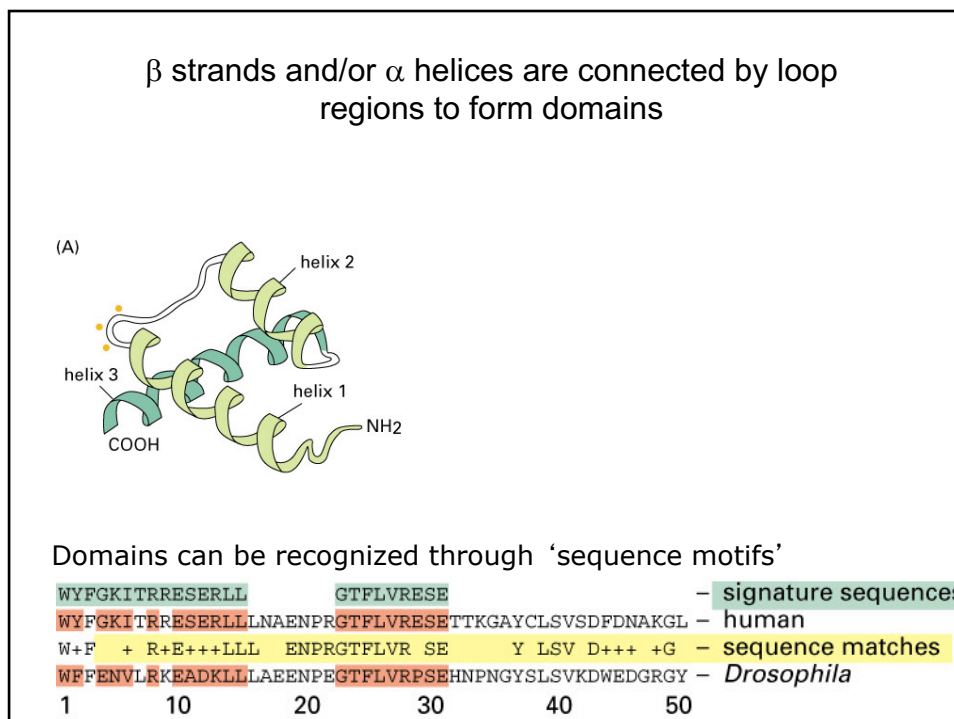
The white dots indicate hydrogen bonds.

Examples of turns

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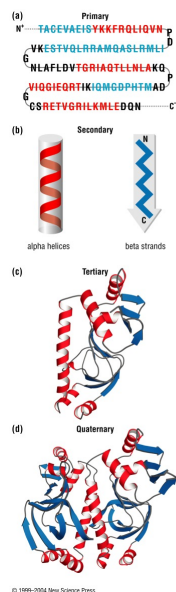
Independent folding domains may have defined functions

Domain	Function
SH2	binds phosphotyrosine
SH3	binds proline-rich sequences
Pleckstrin homology (PH)	binds to G proteins and membranes
WD40	protein–protein interaction
DH	guanine nucleotide exchange
EF-hand	binds calcium
Homeobox	binds DNA
TRBD	binds tRNA
Helix-turn-helix	binds DNA

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Protein folding

From *Protein Structure and Function*
by Gregory A Petsko and Dagmar Ringe



Primary

polypeptide chain

Secondary

alpha-helices,
beta-sheets
turns

Tertiary

domain fold

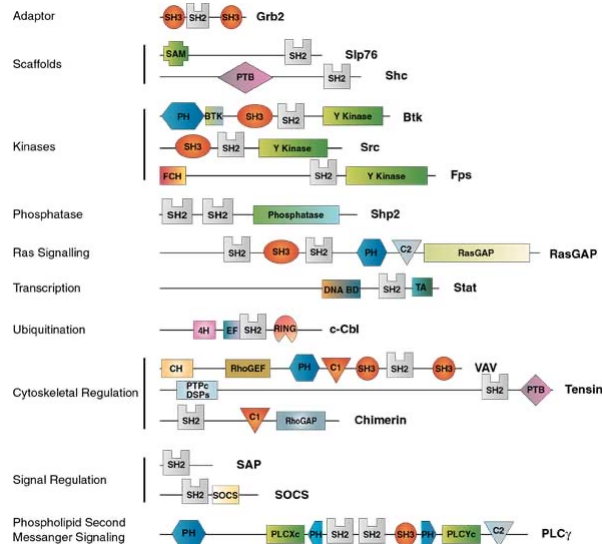


Quaternary

relative orientation
of domains and
subunits

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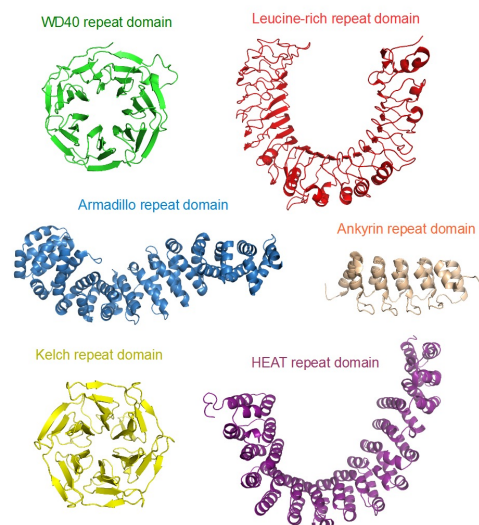
Multiple domains can be mixed and matched in different proteins



Example: SH2 domain proteins, picture: pawson lab

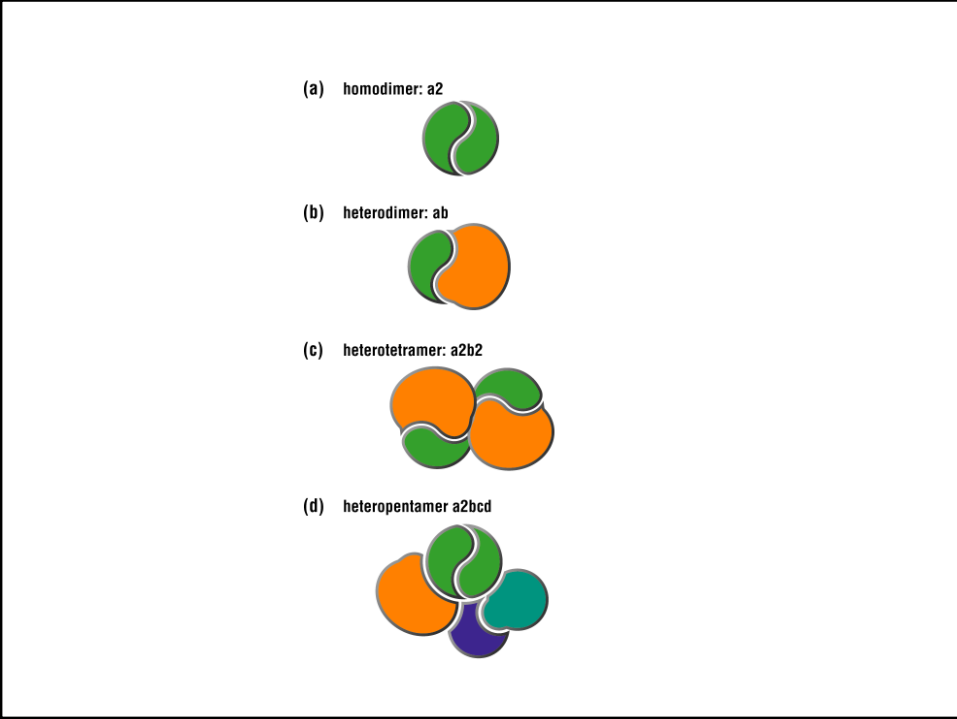
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Large proteins can have a repeat architecture (solenoid)

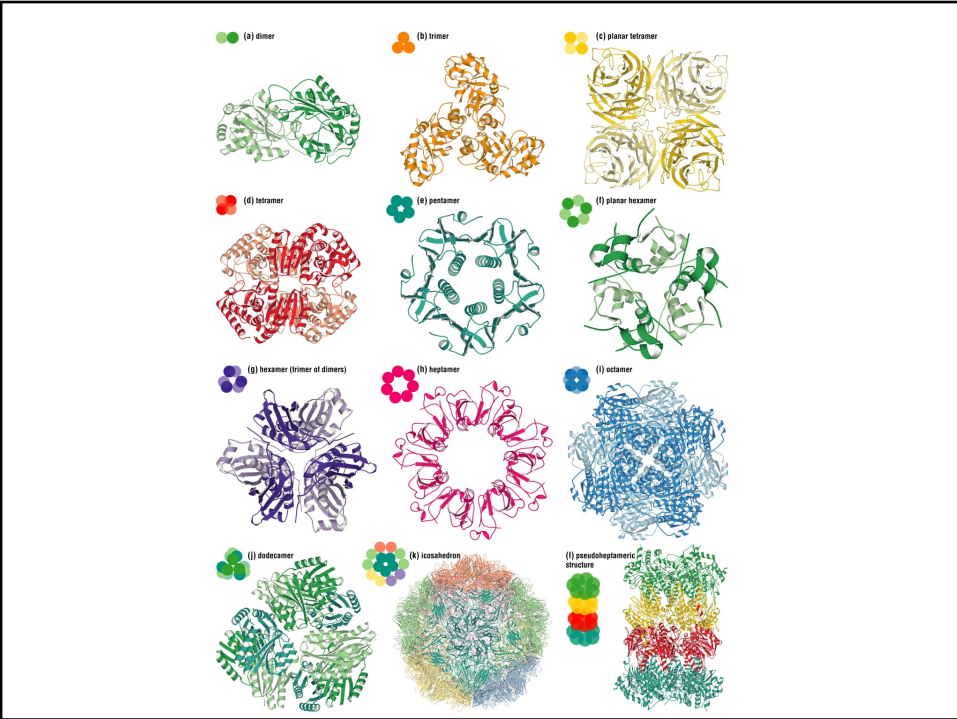


Picture: wikipedia 'solenoid protein domain'

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Post translational modifications (PTMs)

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Post translational modifications (PTMs)

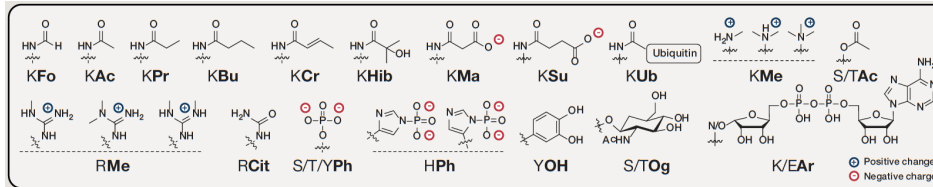
- Disulfide bridge formation
- Cofactor binding
- Proteolysis
- Post-translational modification:
 - Glycosylation Asn, Ser
 - Phosphorylation Tyr, Ser, Thr, His, Asp
 - ADP-ribosylation Arg, Glu, Asp
 - Carbamylation Lys
 - Acetylation Amino groups
 - Methylation Lys, Arg
 - Ubiquitination Lys
 - Lysine hydroxylation Lys

Example: Histone modifications on Chromatin

Challenge: unravel the signals
protein as 'microchip'

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Post-translational histone modifications



Me	Methylation (K, R)
Ac	Acetylation (K, S, T)
Pr	Propionylation (K)
Bu	Butyrylation (K)
Cr	Crotonylation (K)
Hib	2-Hydroxyisobutyrylation (K)
Ma	Malonylation (K)
Su	Succinylation (K)
Fo	Formylation (K)
Ub	Ubiquitination (K)
Cit	Citrullination (R)
Ph	Phosphorylation (S, T, Y, H)
OH	Hydroxylation (Y)
Og	O-GlcNAcylation (S, T)
Ar	ADP ribosylation (K, E)

Huang et al,
Snapshot: histone modifications
Cell 159, October 2014, 458–458.e1

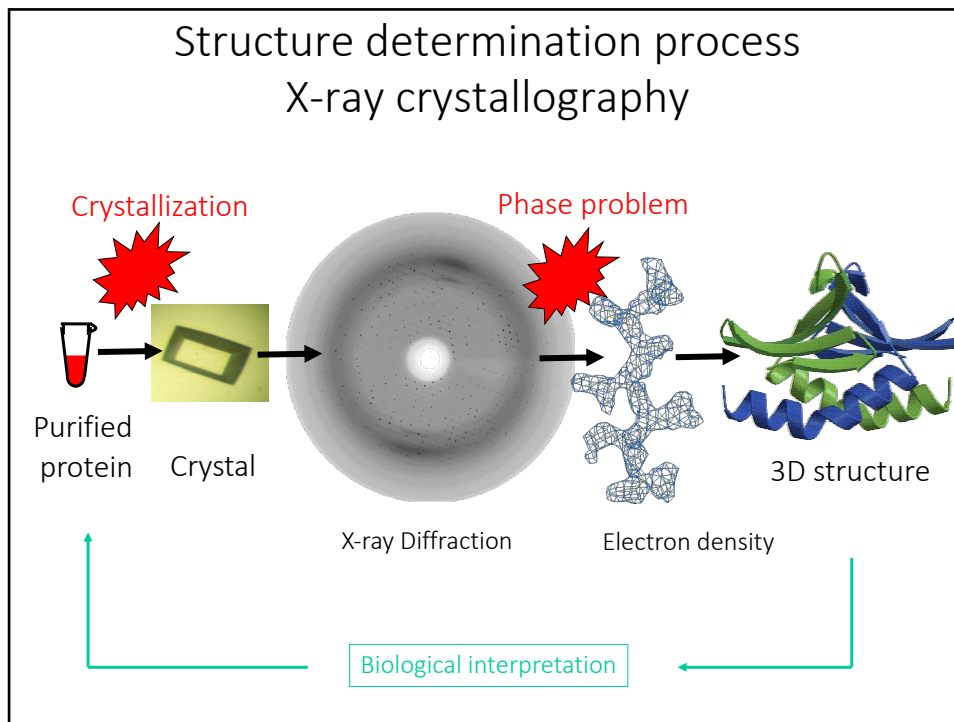
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Intrinsically disordered proteins (IDP)

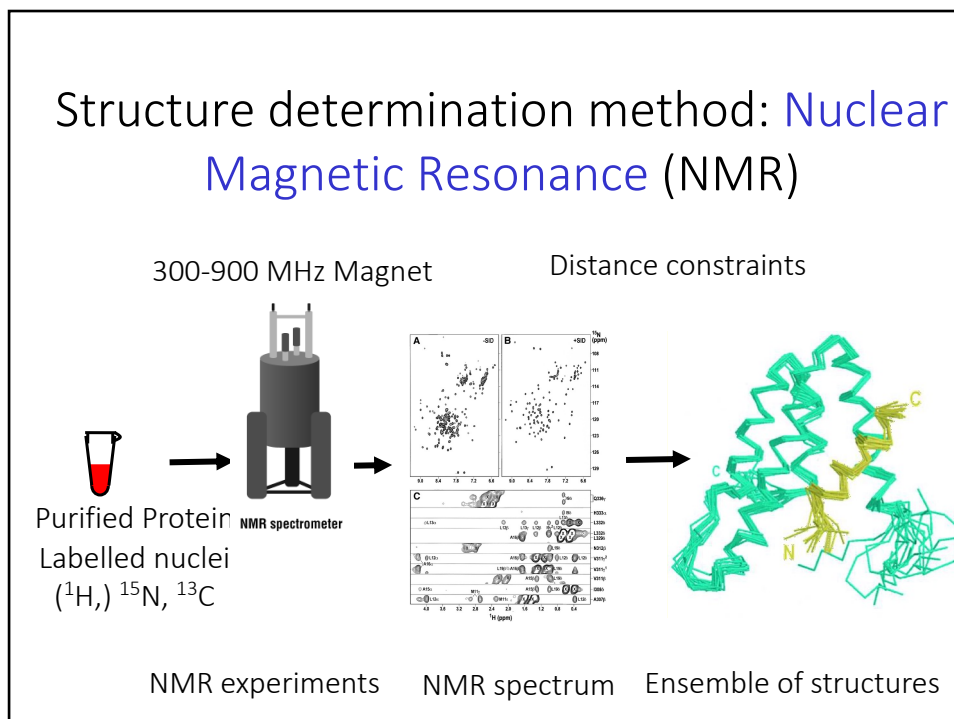
- Many proteins contain natively unfolded regions
 - Flexible linkers
 - Linear motifs
 - Binding regions
 - PTM sites
 - Regions that fold upon interaction
 - coupled folding and binding
 - Regions that remain disordered in the bound state
- Can be stabilized by e.g. chaperones
- Aggregation of IDPs can be functional
 - Method of signalling: Phase separation
 - Basis of disease: Synucleinopathies

» See Theillet et al. 2016

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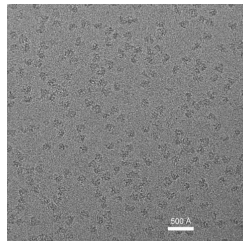
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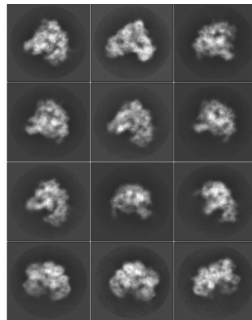
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Single-particle cryo-electron microscopy (EM)

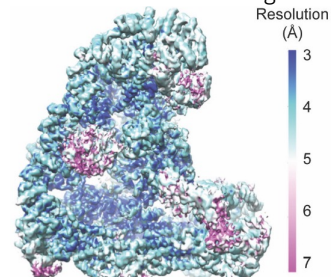
Select particles



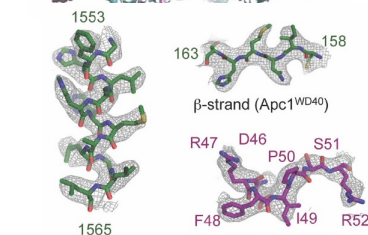
Sort into classes and average



Reconstruct 3D image



2014- today: Revolution in detectors and software is improving resolution to atomic resolution



Fit structural model into data

Nobelprize 2017

Figures from: Chang et al, Nature 2015

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PDB: Protein Data Bank

- **Protein data bank** stores atomic resolution structures of proteins, peptides, viruses, protein-nucleic acids complexes, carbohydrates
 - Xray structures: ± 187.000
 - NMR structures: ± 14.000
 - EM structures: ± 22.000
- **Websites**
 - <http://www.ebi.ac.uk> PDBe
 - <http://www.rcsb.org> RCSB
- **Simple viewers** are available for PC/Mac
 - Astex viewer
 - Pymol (<http://pymol.sourceforge.net/>)
 - SwissPDBviewer and rasmol are older and less powerful
 - CCP4MG
 - ChimeraX (<https://www.cgl.ucsf.edu/chimerax/>)

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A PDB file is a text file

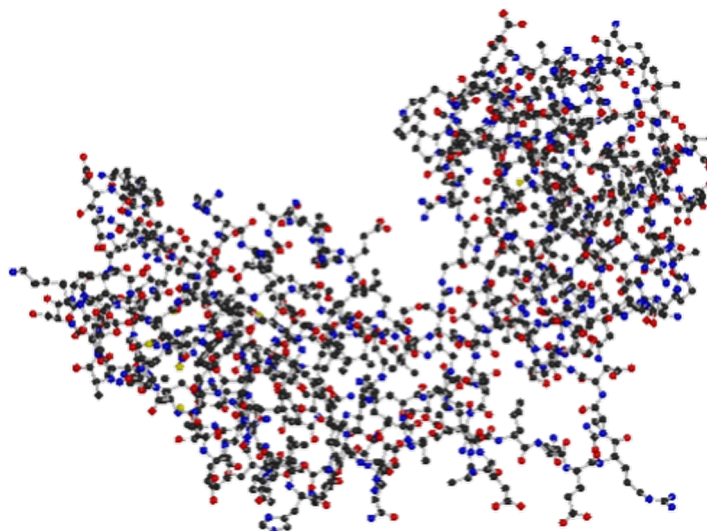
The PDB contains structural data in PDB format or the more structure mmCIF format

The files contain general information on the experiment and the coordinates of the structure, with one line per atom:

	Atom		Residue		Coordinates x,y,z	Thermal motion, B	
ATOM	1	N	SER A	2	-12.168 35.044 19.421	1.00 12.72	N
ATOM	2	CA	SER A	2	-10.726 35.335 19.722	1.00 13.30	C
ATOM	3	C	SER A	2	-10.551 36.793 20.143	1.00 13.69	C
ATOM	4	O	SER A	2	-10.787 37.691 19.352	1.00 13.81	O
ATOM	5	CB	SER A	2	-9.842 35.042 18.500	1.00 13.24	C
ATOM	6	OG	SER A	2	-8.485 35.354 18.766	1.00 11.94	O
ATOM	7	N	ALA A	3	-10.092 37.021 21.367	1.00 14.23	N
ATOM	8	CA	ALA A	3	-10.233 38.325 22.017	1.00 14.64	C
ATOM	9	C	ALA A	3	-9.251 39.384 21.545	1.00 15.04	C
ATOM	10	O	ALA A	3	-8.086 39.091 21.287	1.00 15.17	O
ATOM	11	CB	ALA A	3	-10.109 38.166 23.514	1.00 14.80	C
ATOM	12	N	ILE A	4	-9.742 40.621 21.467	1.00 15.64	N
ATOM	13	CA	ILE A	4	-8.922 41.794 21.169	1.00 16.06	C
ATOM	14	C	ILE A	4	-8.248 42.273 22.455	1.00 16.61	C

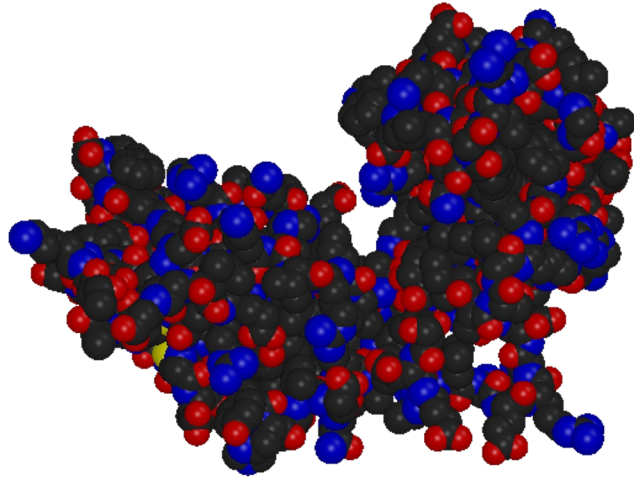
59

Display in ball-and-stick model



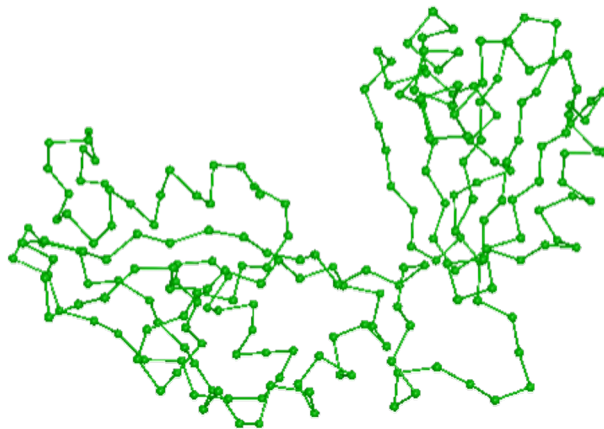
60

Show all atoms



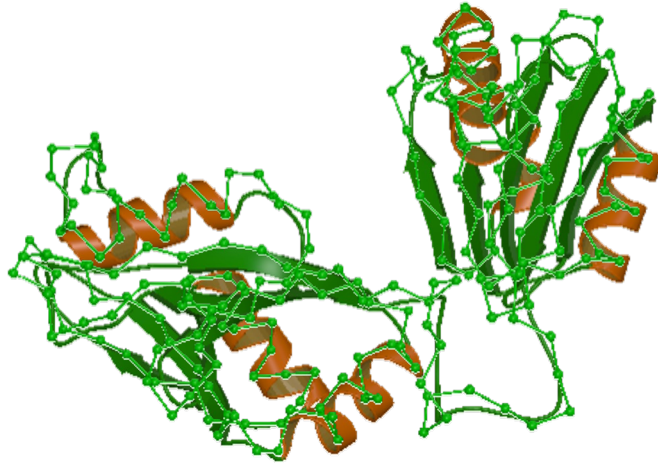
61

Look at the C α trace



62

Accentuate secondary structure



63

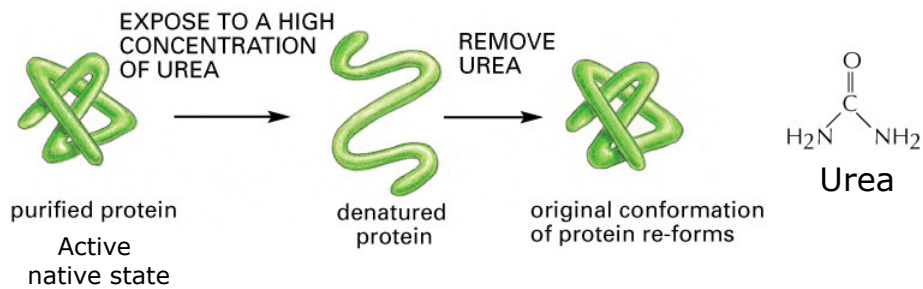
Accentuate secondary structure



64

Anfinsen's principle (1963)

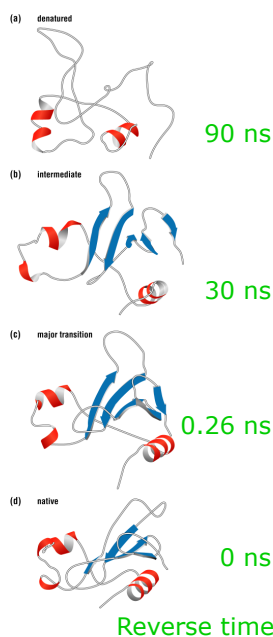
“The information needed to properly fold a protein into its NATIVE 3-D structure is contained in its amino acid sequence”



Primary structure specifies the higher structure

65

The protein folding problem: Levinthal's paradox



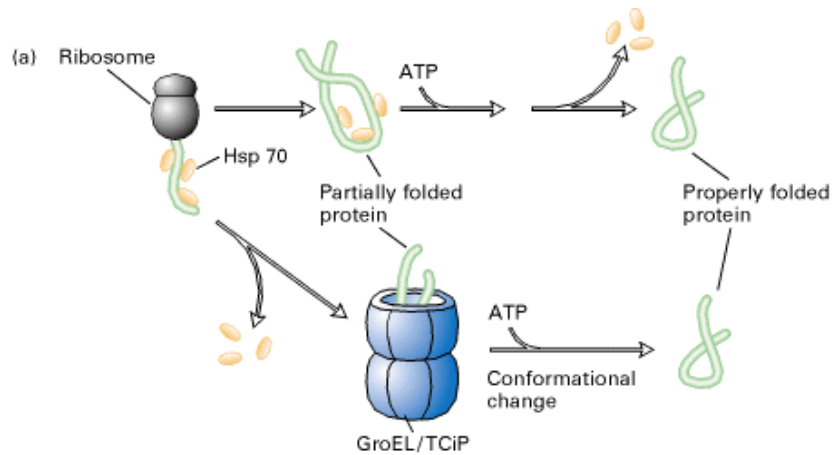
If a small protein folds into its native state by randomly checking all possible conformations of its unfolded state, the process would take longer than the age of the universe (>100 billion years)

Yet, folding can occur in a fraction of a second!

Proteins fold through pathways and mechanisms that remove the need to probe all possible conformations

66

Folding of many proteins in vivo is assisted by chaperones



Chaperones act as catalysts and do not violate the Anfinsen's principle

67

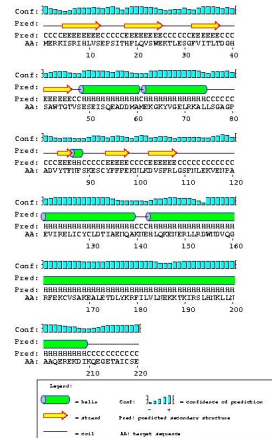
Fold prediction

- Protein structure is more conserved than sequence
 - Until 2021 only:
 - Prediction from other structures and the sequence
 - With very similar proteins: >70% identity
 - Good homology models
 - With similar proteins: >30% identity
 - Good general topology, no detail in insertions/deletions
 - With low sequence identity:
 - Beginning to work: Rosetta@home, massive trials
- Steps:
 - Secondary structure prediction
 - Threading on known folds
 - Analysis of co-evolution

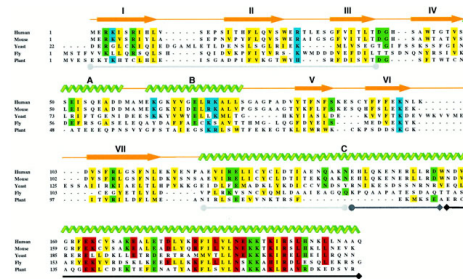
68

Prediction of secondary structure

Predicted



From crystal structure



Prediction of secondary structure is reasonably accurate:

www.expasy.ch or EBI website

Check 'CASP' competition for progress in the prediction field

69

Use of coevolution analysis

If one amino acid change leads to change at a distant site:
infer that they are likely close in space

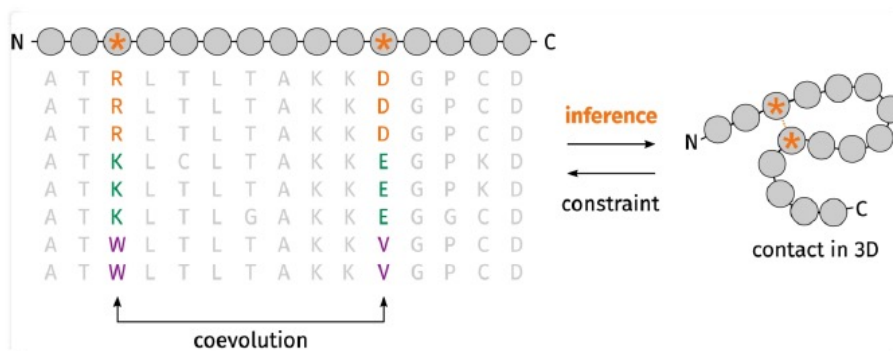


Image: <https://www.bloig.com/blog/2021/07/alphafold-2-is-here-whats-behind-the-structure-prediction-miracle/>

70

The Artificial intelligence watershed moment

- **Alphafold**, an artificial intelligence approach to protein folding results in near-perfect prediction of protein folds.
Announced dec 2020, published Jul 2021, AlphaFold3 may 2024
- Based on multi-sequence alignments, co-evolution analysis and known protein structures from PDB
- Pioneering innovative AI approaches,
- Has now been applied to all proteins in increasing series of genomes: **Alphafold DB** (<https://alphafold.ebi.ac.uk/>)
- Server available for 20 predictions/day

Jumper et al, 2021,
Abramson et al, 2024

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The AlphaFold revolution

- Brings a great new resource of structural information

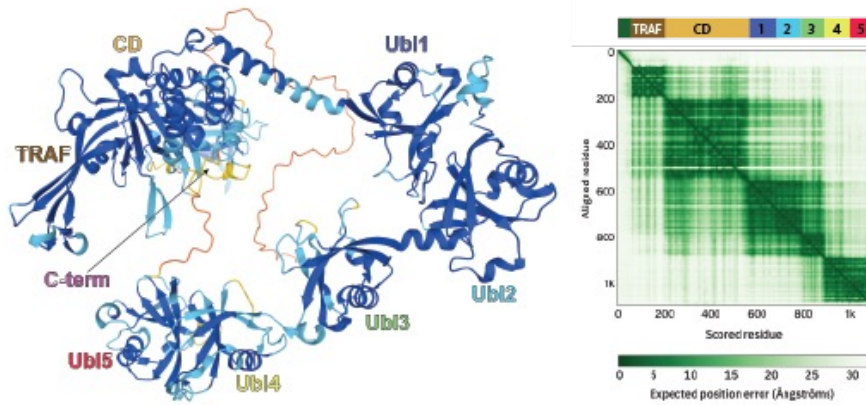
With some limitations

- Sometimes hallucinates (AF3) structure in unfolded regions
- Has only subset of cofactors, metals, ligands and ions
- Only predicts single state for each protein

Nobelprize 2024 in chemistry

72

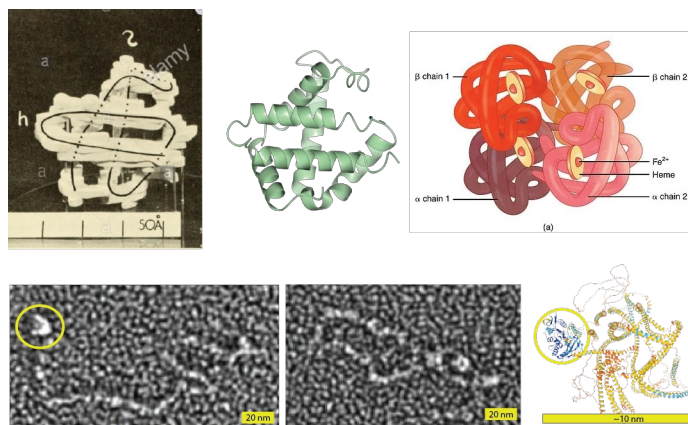
AlphaFold DB



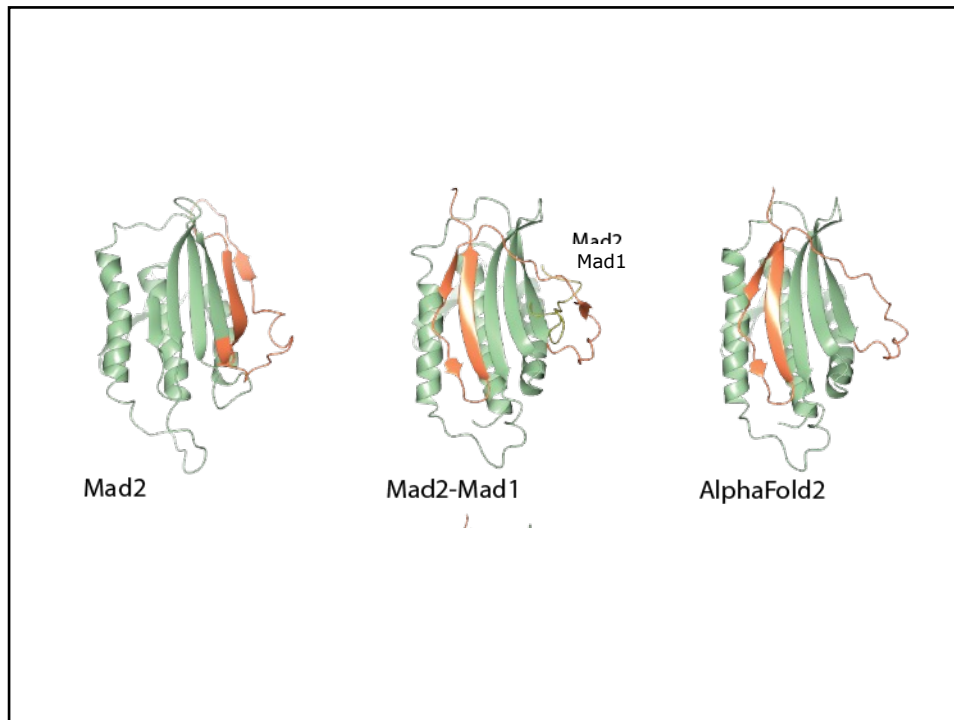
- Colour coding shows confidence in prediction
- Matrix shows confidence in distance between amino acids

73

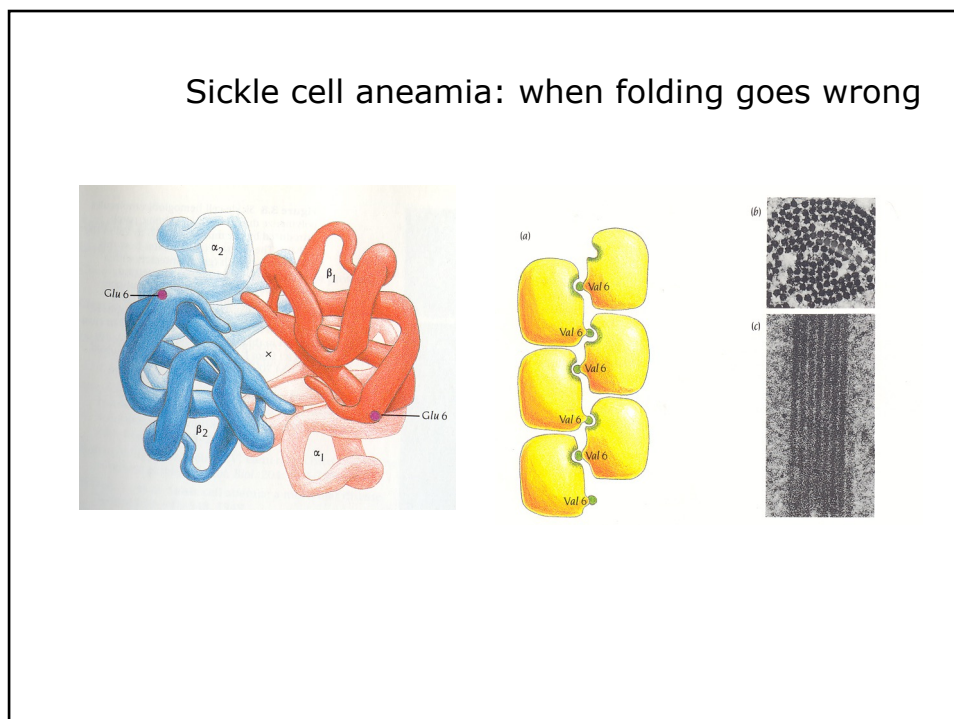
The AlphaFold revolution



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75



76

Static view of protein structure



Dynamic world of protein function

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**Types of Motion Found in Proteins
(all values approximate)**

Motion	Spatial displacement (Å)	Characteristic time (s)	Energy source
Fluctuations (e.g., atomic vibrations)	0.01 to 1	10^{-15} to 10^{-11}	$k_B T$
Collective motions (A) fast, infrequent (e.g., Tyr, Phe ring flips) (B) slow (e.g., domain movement; hinge-bending)	0.01 to > 5	10^{-12} to 10^{-3}	$k_B T$
Triggered conformational changes	0.5 to > 10	10^{-9} to 10^3	Binding interactions

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Protein function

Binding

Specific recognition of other molecules is central to protein function. The molecule that is bound (the ligand) can be as small as the oxygen molecule that coordinates to the heme group of myoglobin, or as large as the specific DNA sequence (called the TATA box) that is bound—and distorted—by the TATA binding protein. Specific binding is governed by shape complementarity and polar interactions such as hydrogen bonding.

TATA binding protein

The TATA binding protein binds a specific DNA sequence and serves as the platform for a complex that initiates transcription of genetic information. (PDB 1tgh)

Myoglobin

Myoglobin binds a molecule of oxygen reversibly to the iron atom in its heme group (shown in grey with the iron in green). It stores oxygen for use in muscle tissues. (PDB 1t8k)

Catalysis

Essentially every chemical reaction in the living cell is catalyzed, and most of the catalysts are protein enzymes. The catalytic efficiency of enzymes is remarkable: reactions can be accelerated by as much as 17 orders of magnitude over simple buffer catalysis. Many structural features contribute to the catalytic power of enzymes: holding reacting groups together in an orientation favorable for reaction (proximity), bending the transition state of the reaction more tightly than ground state complexes (transition state stabilization), acid-base catalysis, and so on.

DNA polymerase

DNA replication is catalyzed by a specific polymerase that copies the genetic material and edits the product for errors in the copy. (PDB 1pdx)

HIV protease

Replication of the AIDS virus HIV depends on the action of a protein-cleaving enzyme called HIV protease. This enzyme is the target for protease-inhibitor drugs (shown in grey). (PDB 1atg)

Switching

Proteins are flexible molecules and their conformation can change in response to changes in pH or ligand binding. Such changes can be used as molecular switches to control cellular processes. One example, which is critically important for the molecular basis of many cancers, is the conformational change that occurs in the small GTPase Ras when GTP is hydrolyzed to GDP. The GTP-bound conformation is an "on" state that signals cell growth; the GDP-bound structure is the "off" signal.

'off'

Ras

'on'

The GDP-bound ("off") PDB 1tgh state of Ras differs significantly from the GTP-bound ("on") PDB 127g) state. This difference causes the two states to be recognized by different proteins in signal transduction pathways.

Structural

Protein molecules serve as some of the major structural elements of living systems. This function depends on specific association of protein subunits with themselves as well as with other proteins, carbohydrates, and so on, resulting even complex systems like actin filaments to assemble spontaneously. Structural proteins are also important sources of biomaterials, such as silk, collagen, and keratin.

Silk

Silk derives its strength and flexibility from its structure. It is a giant stack of antiparallel beta sheets. Its strength comes from the covalent and hydrogen bonds within each sheet; the flexibility from the "van der Waals" interactions that hold the sheets together. (PDB 1s1k)

F-actin

Actin filaments are important for muscle contraction and for the cytoskeleton. They are hollow assemblies of actin and other associated proteins. (Courtesy of Ken Holmes)

Mechanical work

© 1999-2004 New Science Press

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Protein functions

Binding

Bind a ligand,DNA,protein

Catalysis

Promote a reaction

Switchin

Create a signal

Structural

Provide stability

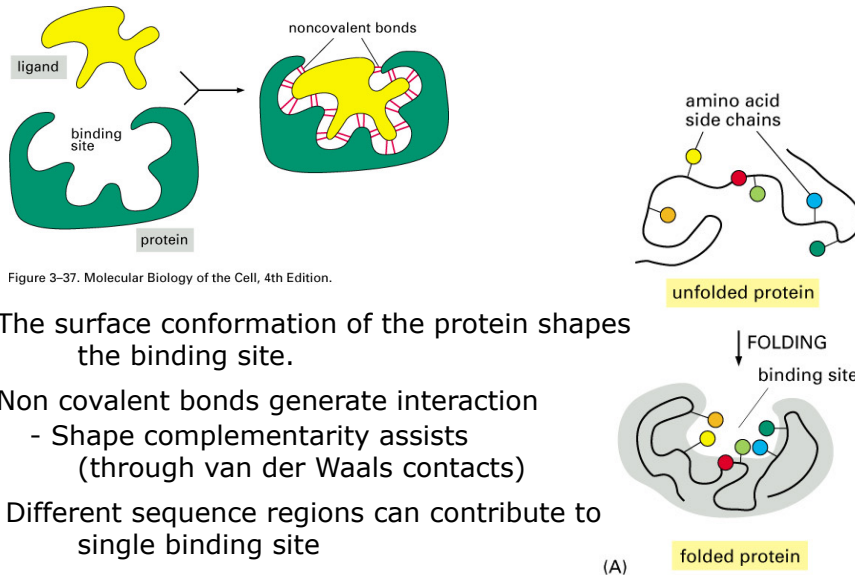
Mechanical work

Transfer energy into motion

80

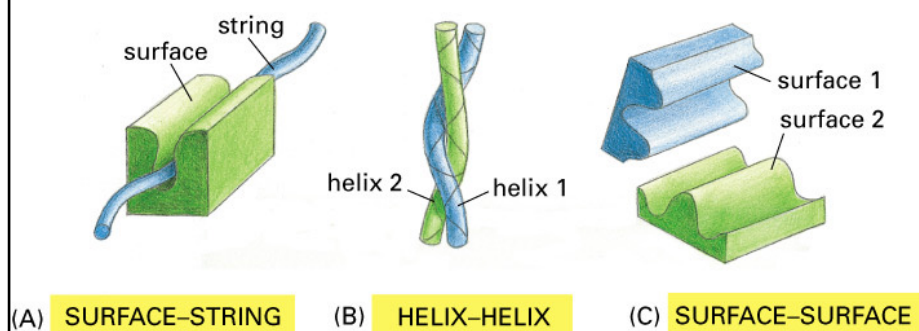
40

Ligand binding



81

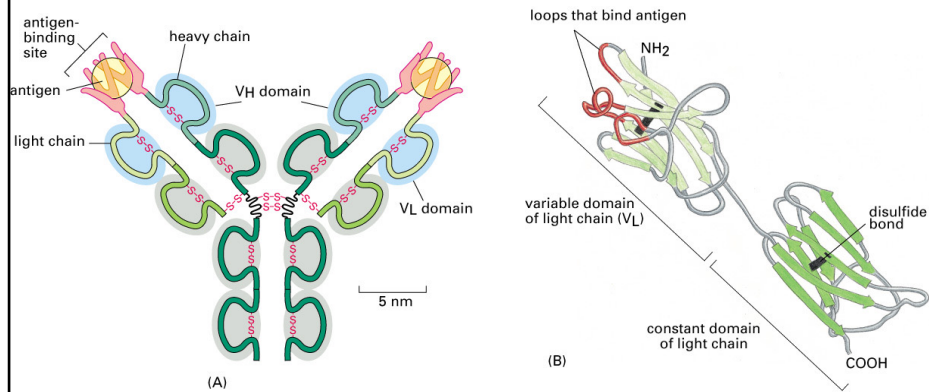
Protein/protein binding



- Proteins bind to other proteins through several types of interfaces

82

Antibodies display versatile protein interactions



83

Binding function of proteins Depends on noncovalent interactions

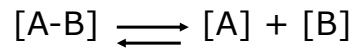
How strong is the affinity? K_b

How fast is the association? k_{on}

How fast is the dissociation? k_{off}

Binding	Bind a ligand, DNA, protein
Catalysis	Promote a reaction
Switchin	Create a signal
Structural	Provide stability
Mechanical work	Transfer energy into motion

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At equilibrium

$$K_D = \frac{[A][B]}{[A-B]}$$

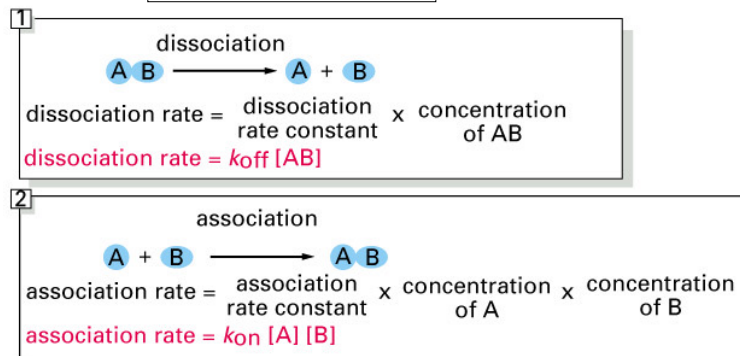
Unit M

Affinity $\nearrow$ when $K_D \searrow$

		Ubiquitin binding domain 10^{-5} M
		Transcription factor 10^{-9} M
Weak	μ M	Antigen-receptor 10^{-7} to 10^{-11} M
Strong	nM	Avidin-biotin 10^{-15} M

85

Binding kinetics



Simple 1 to 1 binding

Give examples where binding kinetics do not obey simple 1 to 1 binding model...

86

Real-time kinetics of binding and dissociation by surface plasmon resonance

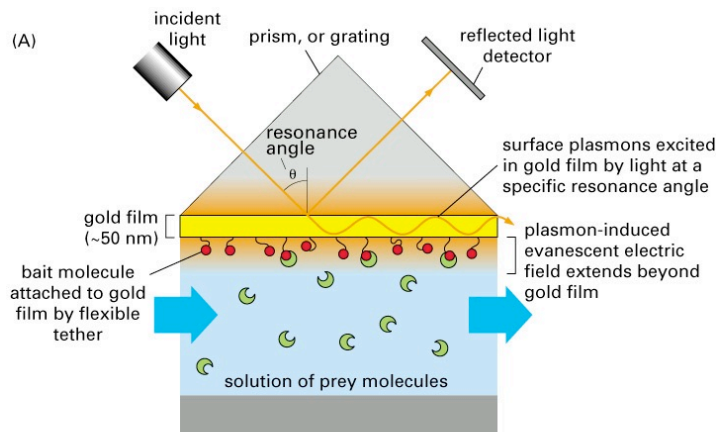


Figure 8-53 part 1 of 2. Molecular Biology of the Cell, 4th Edition.

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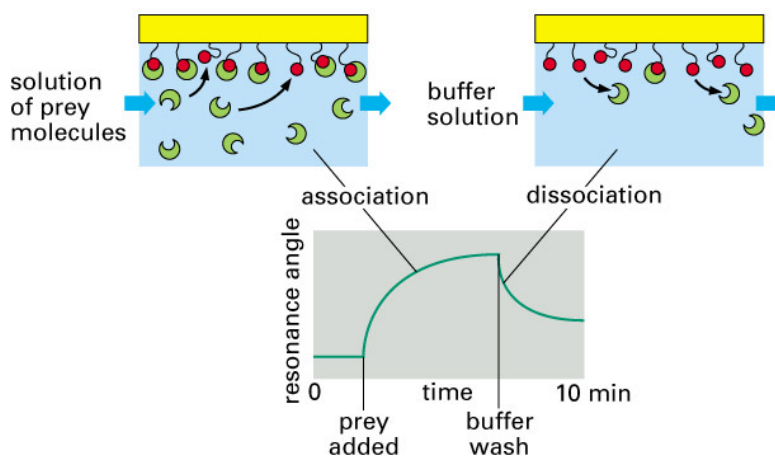


Figure 8-53 part 2 of 2. Molecular Biology of the Cell, 4th Edition.

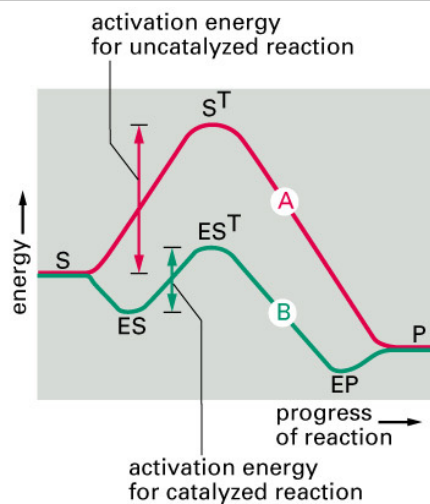
88

Most biochemical reactions are accelerated by protein catalysts

Binding	Bind a ligand, DNA, protein
Catalysis	Promote a reaction
Switching	Create a signal
Structural	Provide stability
Mechanical work	Transfer energy into motion

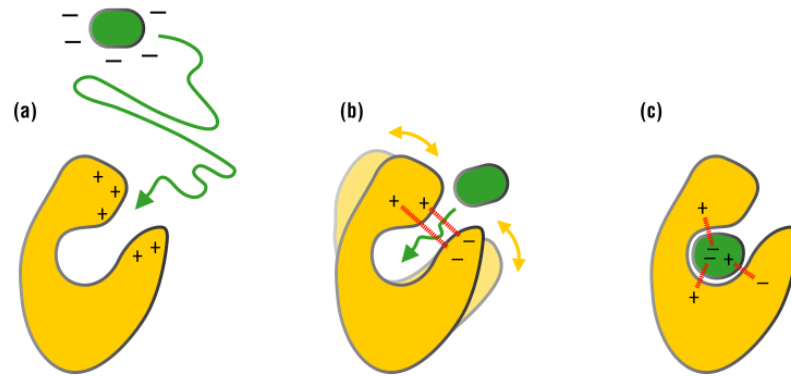
89

Enzymes stabilize the transition state and lower the activation energy barrier



90

The catalytic pocket



91

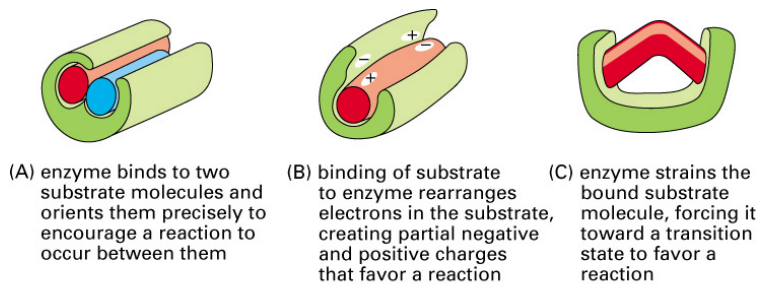
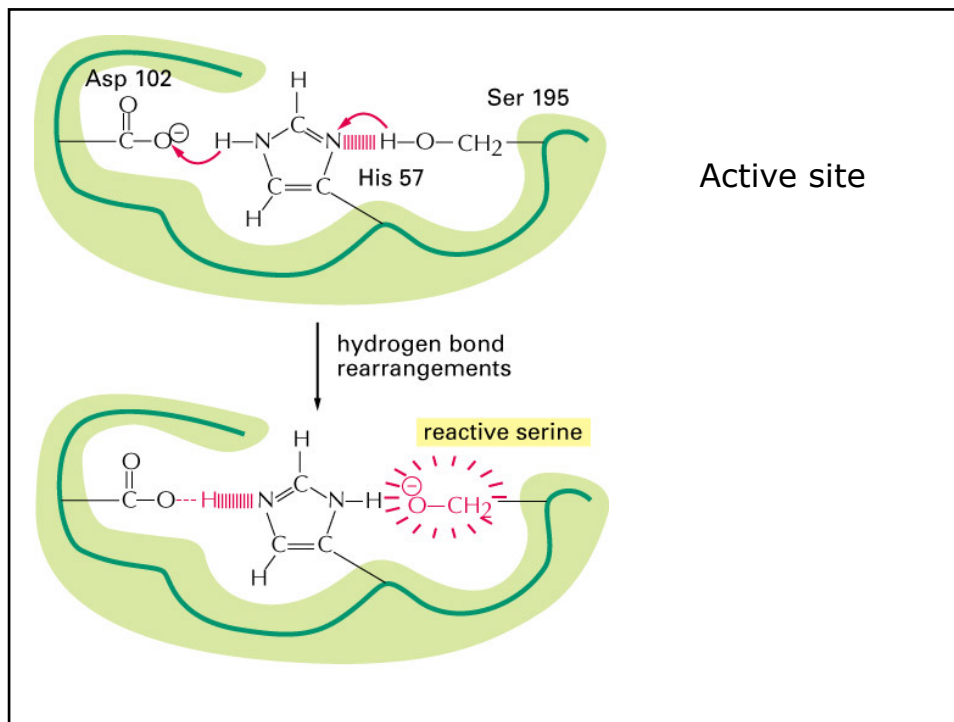


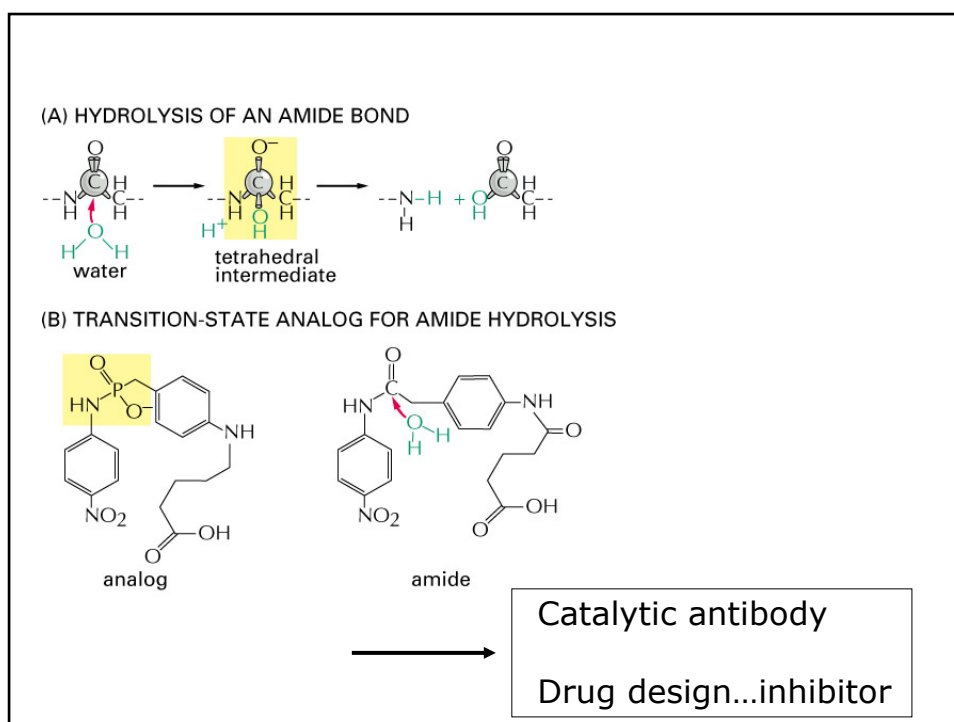
Figure 3-52. Molecular Biology of the Cell, 4th Edition.

- a) Ideal positioning of reactants for reaction
- b) Arrange charges to promote reaction
- c) Strain the substrate to promote reaction

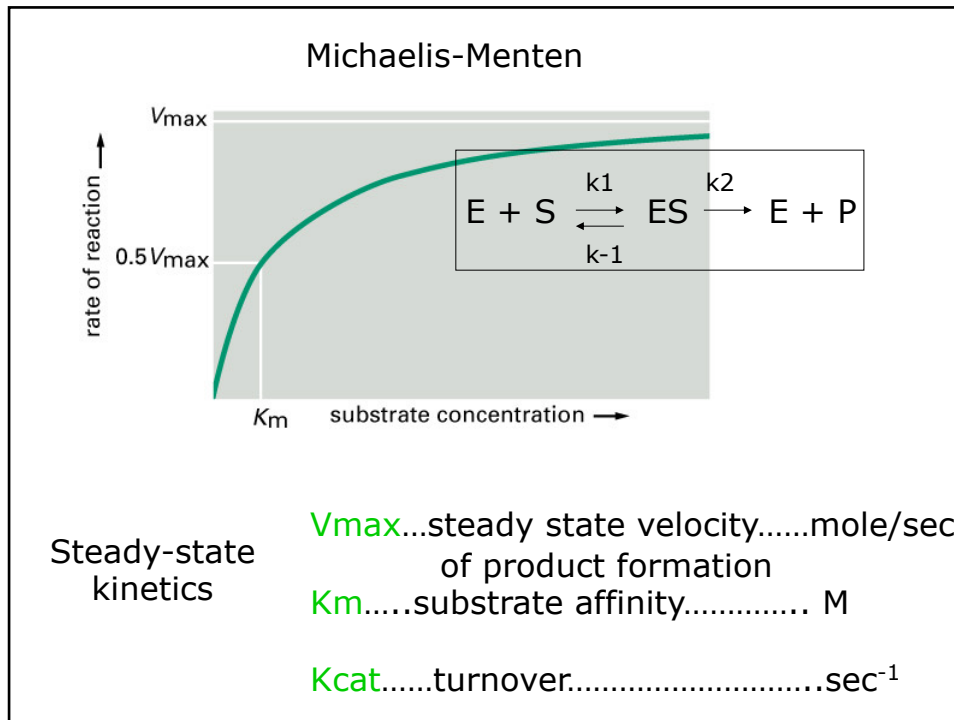
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94



95

Comparison of Uncatalyzed and Catalyzed Rates for Some Enzymatic Reactions			
Enzyme	Nonenzymatic rate $k_{\text{non}} \text{ (s}^{-1}\text{)}$	Enzymatic rate $k_{\text{cat}} \text{ (s}^{-1}\text{)}$	Rate acceleration $k_{\text{cat}}/k_{\text{non}}$
Cyclophilin	2.8×10^{-2}	1.3×10^4	4.6×10^5
Carbonic anhydrase	1.3×10^{-1}	10^6	7.7×10^6
Chymotrypsin	4×10^{-9}	4×10^{-2}	10^7
Triosephosphate isomerase	6×10^{-7}	2×10^3	3×10^9
Fumarase	2×10^{-8}	2×10^3	10^{11}
Adenosine deaminase	1.8×10^{10}	370	2.1×10^{12}
Urease	3×10^{-10}	3×10^4	10^{14}
Alkaline phosphatase	10^{-15}	10^2	10^{17}
ODCase	2.8×10^{-16}	39	1.4×10^{17}

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The cell regulates the catalytic activities of its enzymes

Activation by allosteric binding of other molecules
e.g. aspartate transcarbamoylase

Activation by post translational modification
e.g. phosphorylation

Activation by proteolysis

e.g. blood clotting enzymes, complement

Activation by protein interaction

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Allosteric effects on enzyme activity

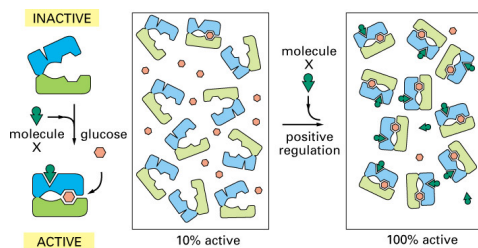


Figure 3-57. Molecular Biology of the Cell, 4th Edition.

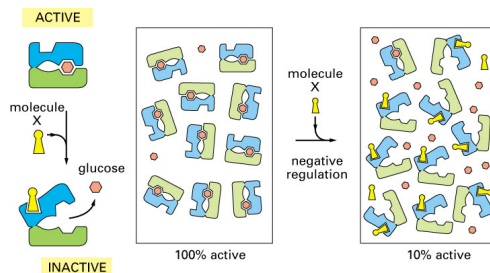


Figure 3-58. Molecular Biology of the Cell, 4th Edition.

- Allosteric enzymes have two or more binding sites that interact
- Two ligands whose binding sites are coupled must reciprocally affect each other's binding
- Activators and inhibitors can use the same or different allosteric sites

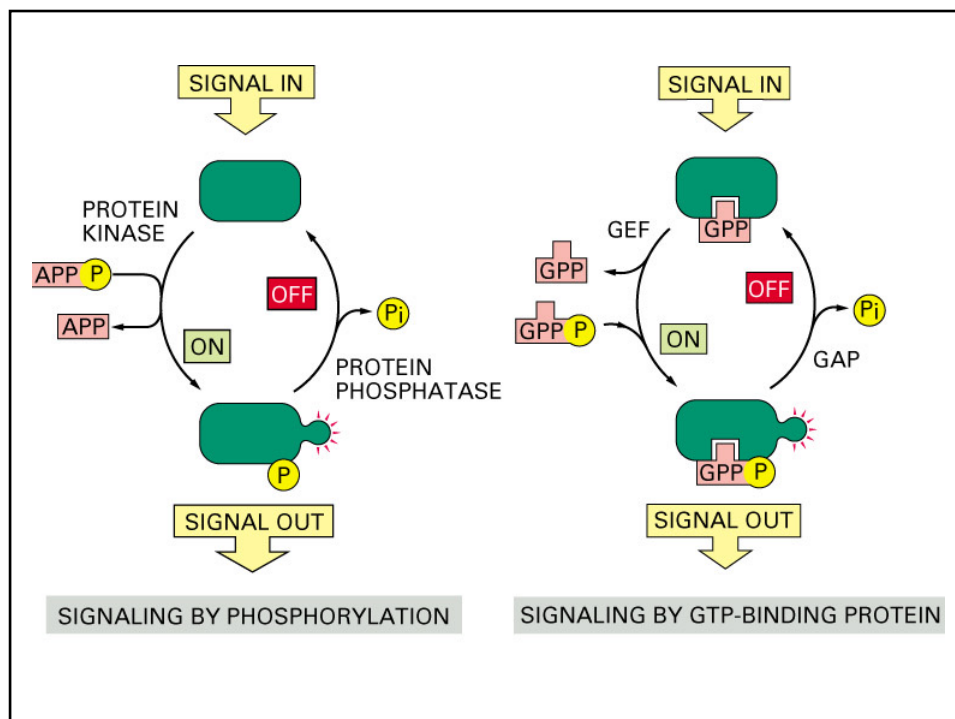
98

Many proteins function as molecular switches

Conformational changes

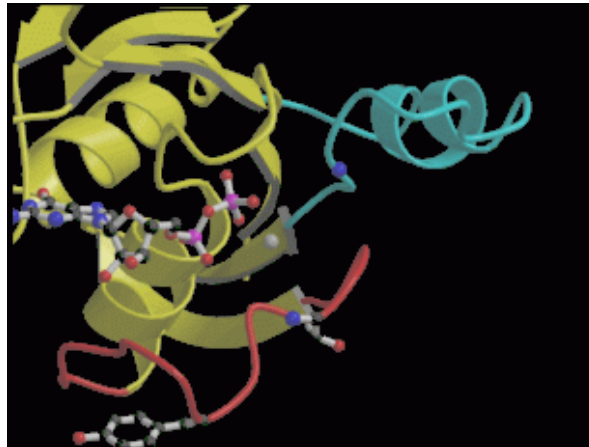
Binding	Bind a ligand,DNA,protein
Catalysis	Promote a reaction
Switching	Create a signal
Structural	Provide stability
Mechanical work	Transfer energy into motion

99



100

Ras GTP/GDP



101

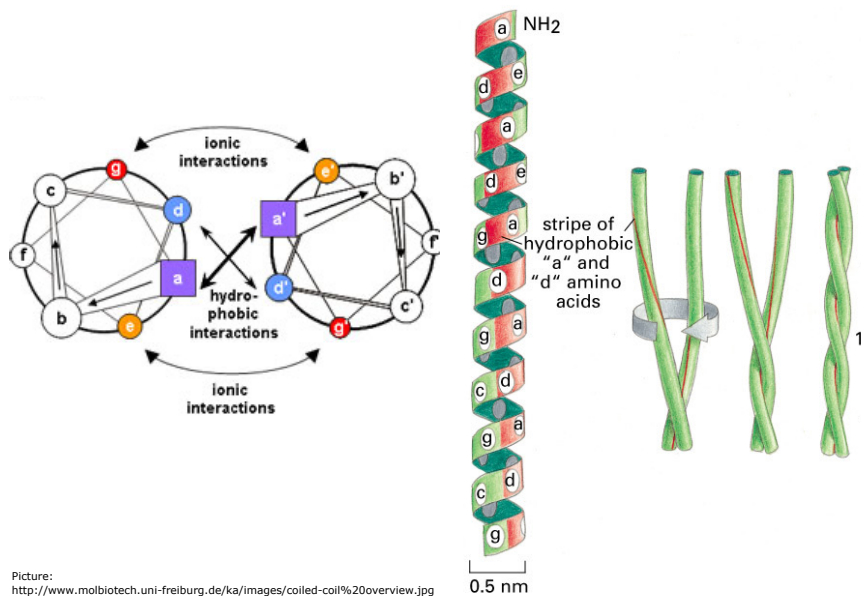
Proteins can provide structure to a cell

Cytoskeleton
Keratin

Binding	Bind a ligand,DNA,protein
Catalysis	Promote a reaction
Switching	Create a signal
Structural	Provide stability
Mechanical work	Transfer energy into motion

102

Coiled coil structures are built from two helices with a regular hydrophobic sequence pattern



103

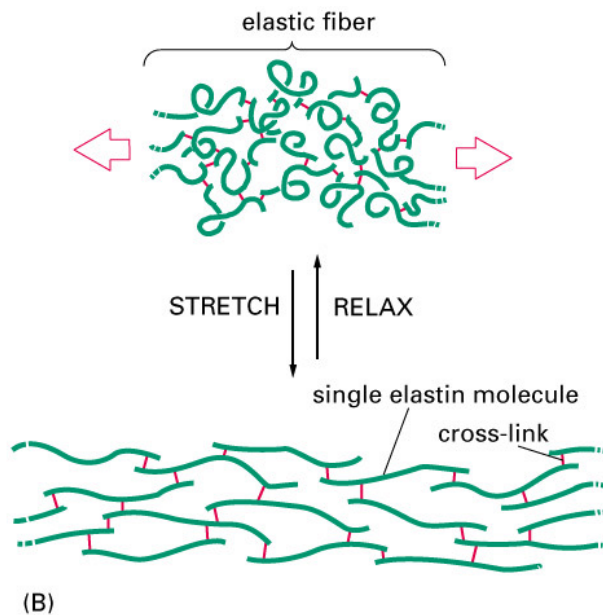


Figure 3-28 part 2 of 2. Molecular Biology of the Cell, 4th Edition.

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Motor proteins use the energy of ATP to generate large movements

Normal conformational states
nondirectional

Binding
Catalysis
Switching
Structural
Mechanical work

Bind a ligand, DNA, protein
Promote a reaction
Create a signal
Provide stability
Transfer energy into motion

105

Mechanical work

Coupling movement to a irreversible reaction

ATP BINDING

HYDROLYSIS

RELEASE

ADP

direction of movement

Myosin on actin filaments

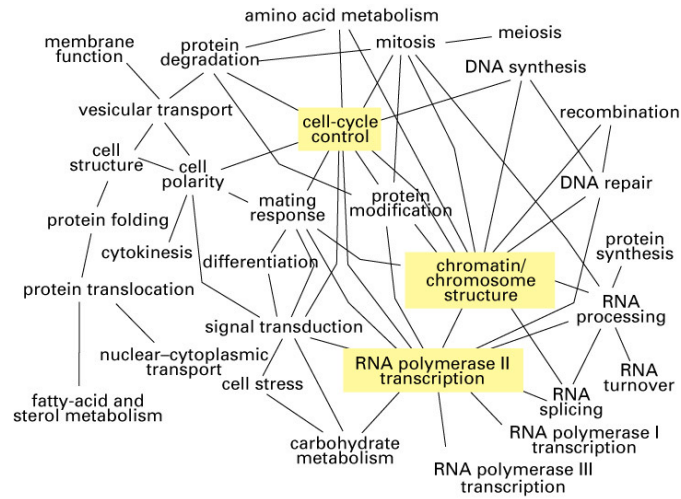
Kinesin on microtubules

Helicase on DNA

Challenge: unravelling steps in multicomponent systems
single molecule, quantitative data, cellular data

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Systems Biology: understanding integration between networks of proteins



Challenge: quantitative data for individual components
Kinetic, Binding